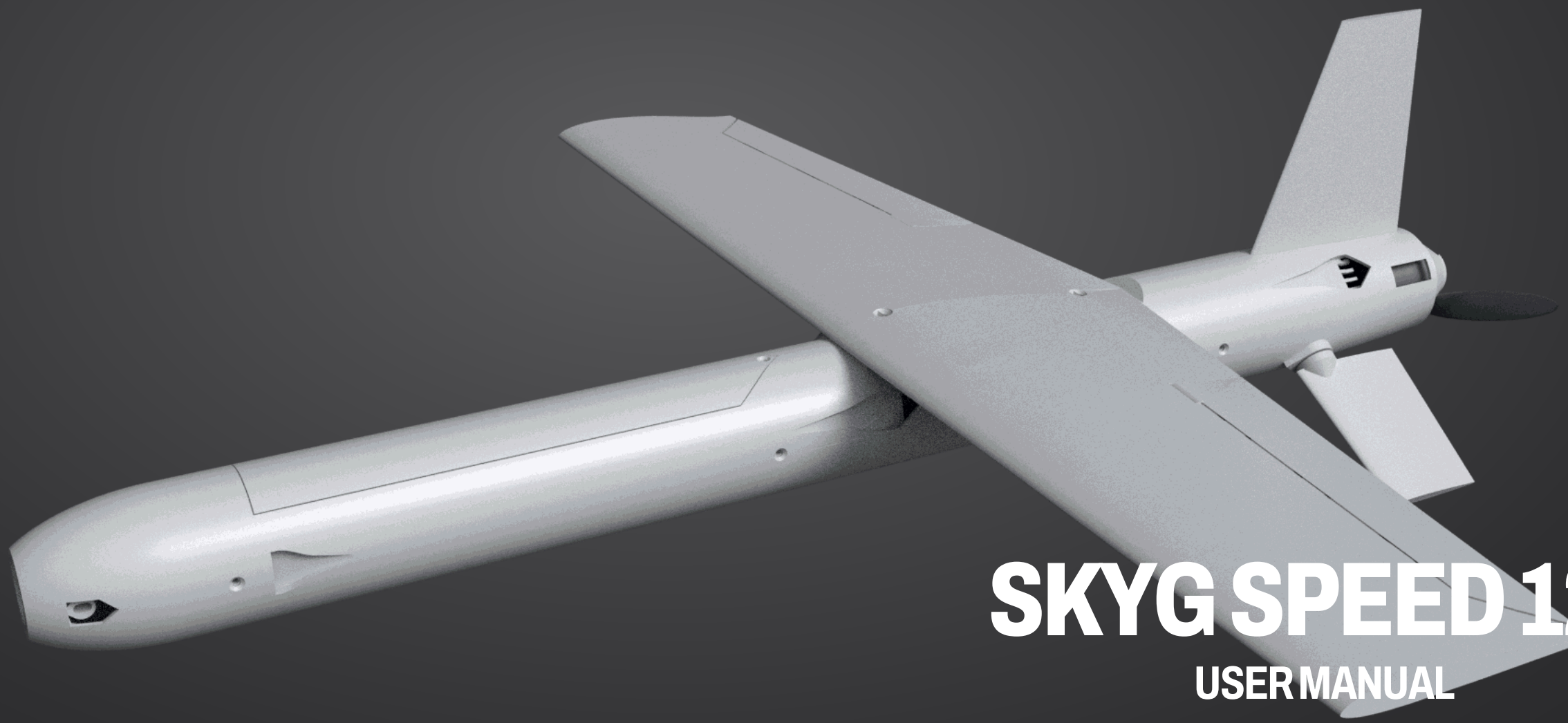
SKYG



SKYG SPEED 120

USER MANUAL

V.1 · C7-2 · 2026-10-02

PACKAGE CONTENTS

Manual

SkyG Speed 120 user manual (this document).

STL

Every printed part, in millimetres, ready for the slicer (print orientation: pages in this manual).

STEP

Every printed part and the full assembly, for editing in any CAD program.

Wiring

The wiring diagram as a separate large page.

ArduPilot

Setup document and a starting .param file for the Matek H743-WLITE, and the rpm-governor script.

Viewer

Interactive 3D model (open the HTML file in a browser): every part exploded and in place, with notes.

Engineering

Design record, parts list (BOM.csv), test plan and verification report.

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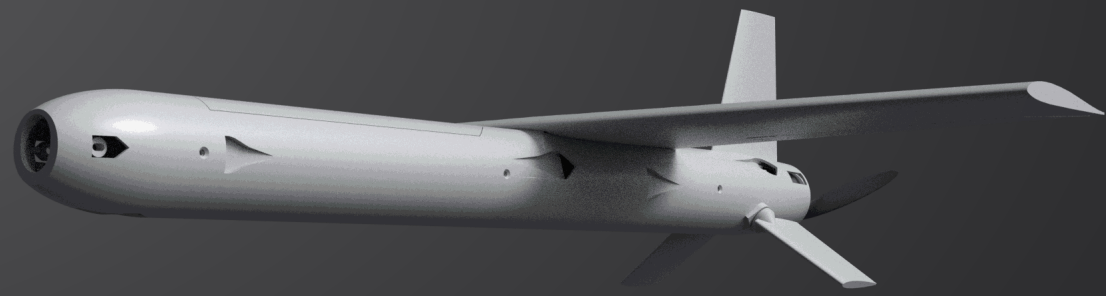
Flight test and speed stages 49

SKYG SPEED 120

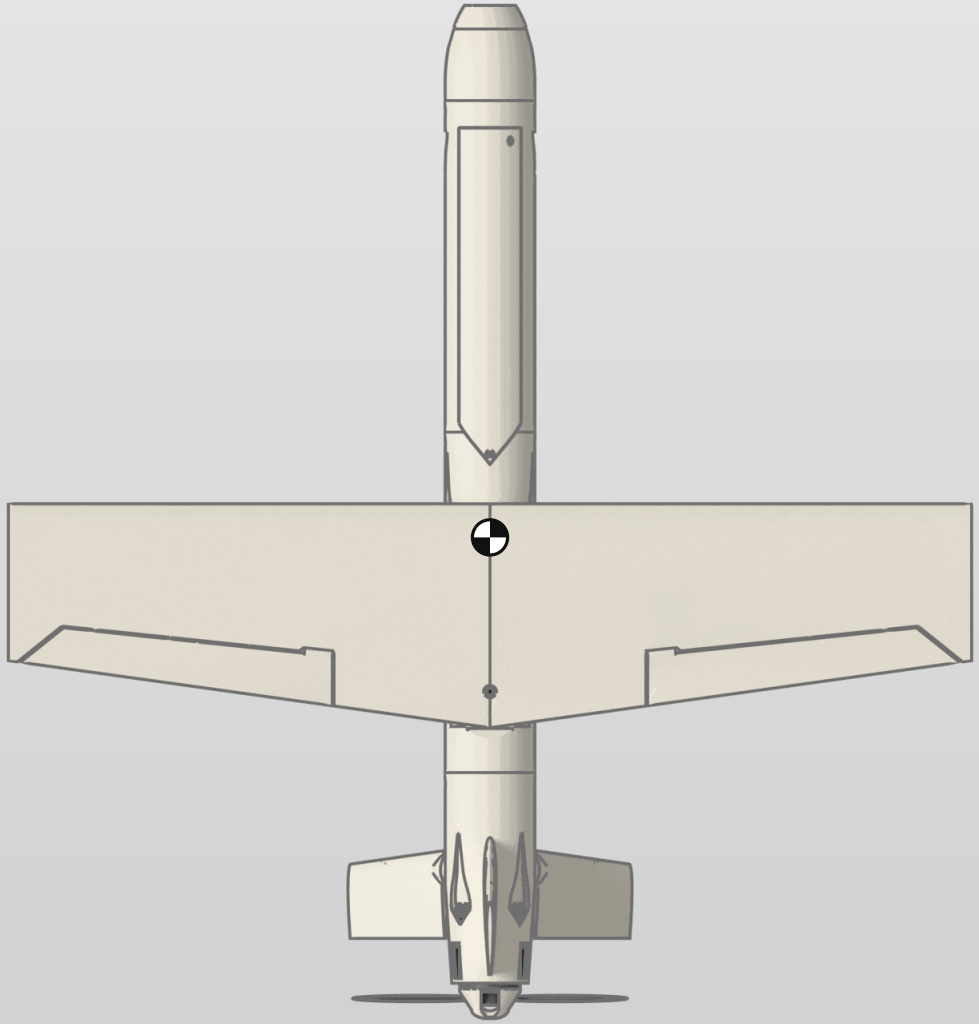
SkyG Speed 120 is a compact, hand-launched speed aircraft: a 3D-printed fuselage in three plug-together modules and a separate camera nose, a 430 mm straight wing on a carbon spar, and an all-moving canted tail driven by two micro servos. A single pusher motor sits in the tail cone behind the tail.

The airframe is printed in foaming ASA Aero on a 0.4 mm nozzle; steel and carbon parts carry the concentrated loads (spar, aileron hinge pins and torque rods, tail shafts). Six interchangeable camera noses take a micro FPV camera, a Raspberry Pi camera or a DJI O4 Air Unit, each with an open or a clear-dome face.

It is built for straight-line speed: about 118 m/s as printed and 120 m/s finished in the model with the clear-dome nose, a 6S pack and an rpm-governed carbon prop, flown up through never-exceed stages to 120 m/s.



GENERAL AIRCRAFT DATA



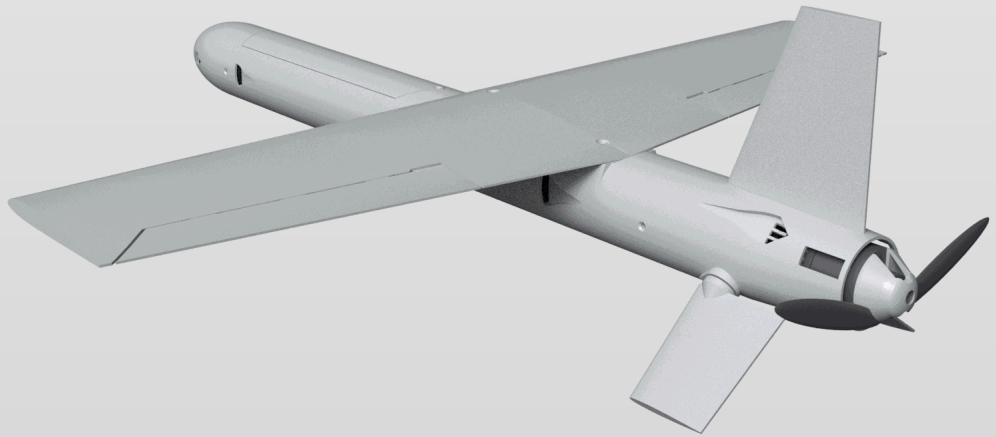
Wingspan	430 mm
Wing area	3.65 dm²
Length	449 mm
Centre of gravity	15 mm behind the wing leading edge at the root
All-up weight	400 g (clear-dome nose)
Stall speed	13.2 m/s
Top speed (model)	118–120 m/s
Airfoil	MH45, ailerons 3° reflex
Root / tip chord	100 / 70 mm
MAC	86 mm
Aspect ratio	5.1
Wing loading	107 g/dm²

AERODYNAMIC DESIGN

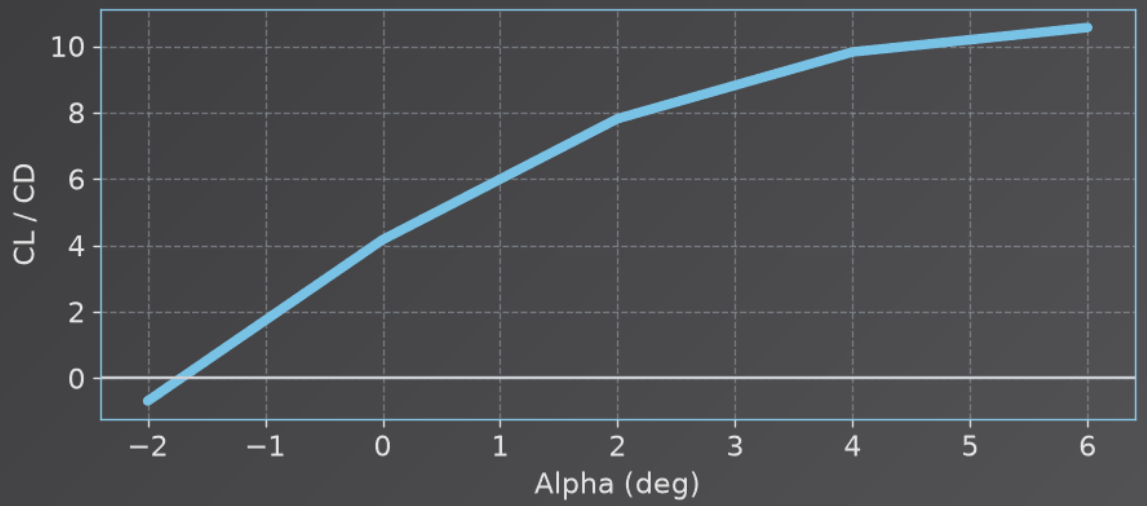
The wing uses the MH45 section, a low-moment reflexed profile, with both ailerons rigged 3° trailing edge up. The design CG is at 10 % of the mean chord, 8.9 % static margin ahead of the neutral point (x 246 mm from the nose datum): stable without a tail download penalty at speed.

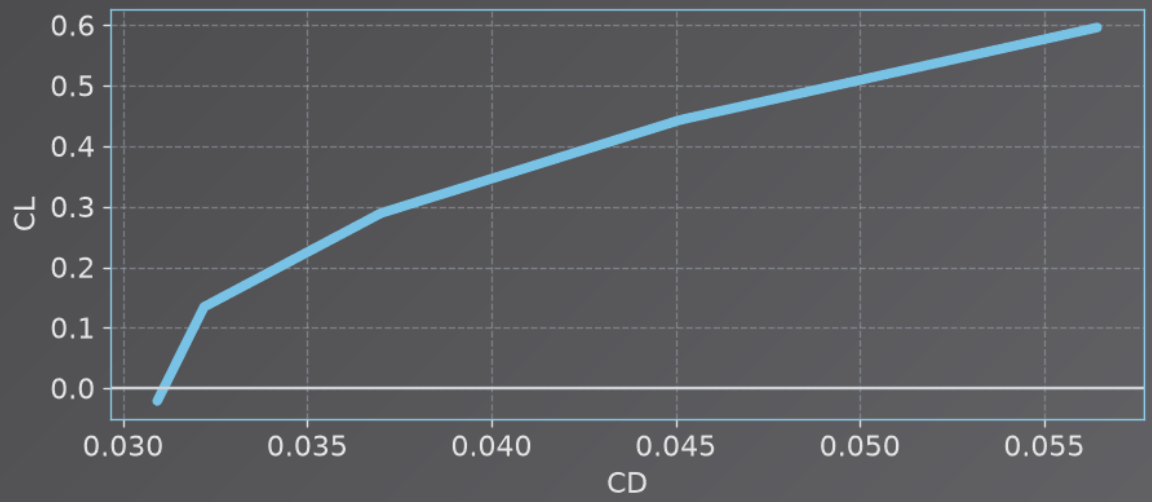
Pitch and yaw come from two all-moving tail panels at 35° anhedral, mixed like a V-tail, with a fixed fin on top. The Dutch roll is lightly damped at low speed, so the flight controller's yaw damper is part of the setup.

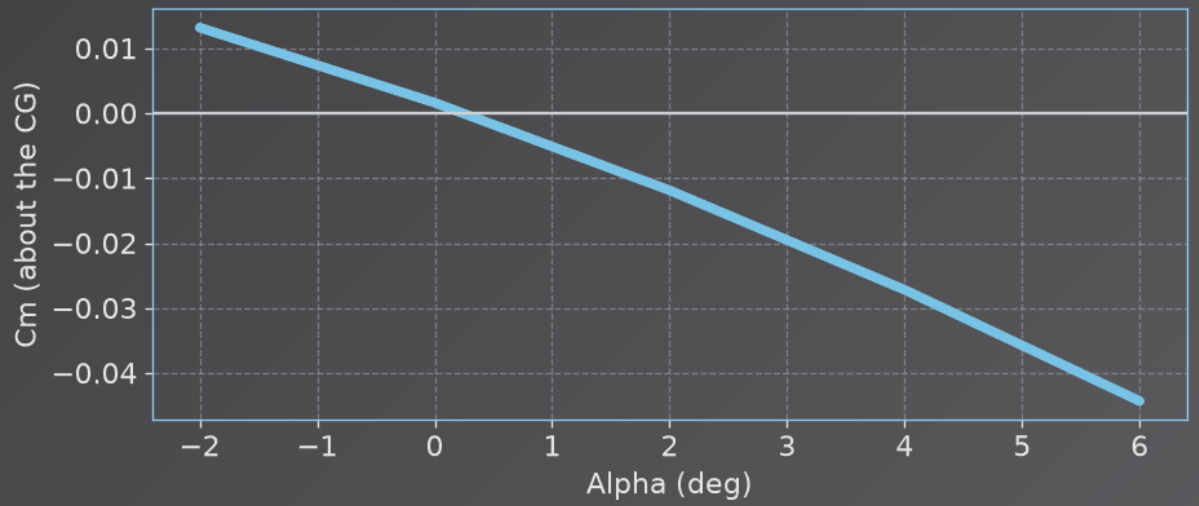
Flutter of the wing, ailerons and tail panels was analysed (strip theory with the CAD-derived stiffness and masses): the lowest predicted instability is 168 m/s, above 1.2 × the 120 m/s top stage. The test plan opens the speed envelope in stages and measures the stiffness the margin depends on.

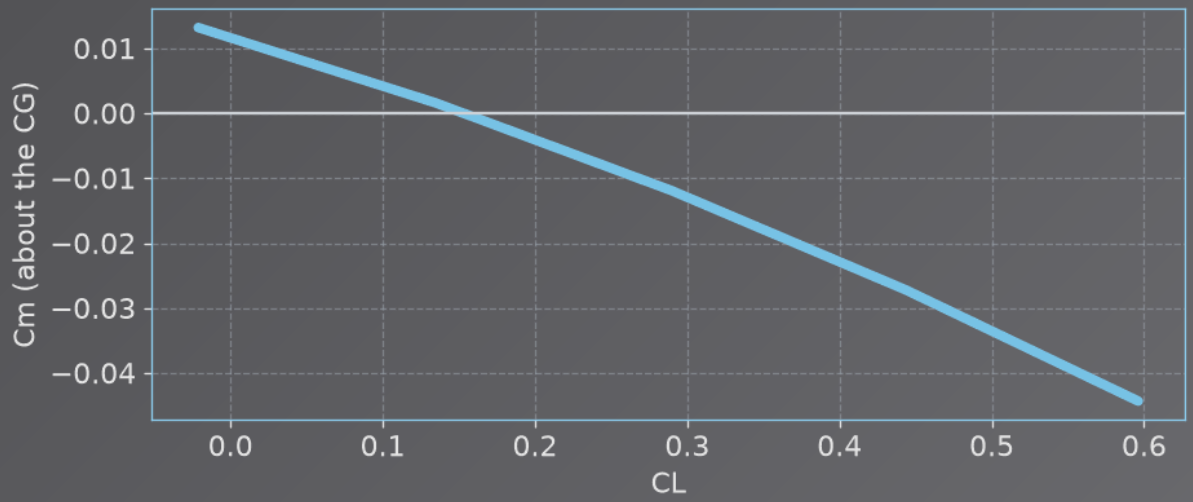


AERODYNAMIC CHARACTERISTICS









Whole aircraft, VSPAERO vortex lattice with the fuselage panels; best L/D 10.6 at 6.0°; Cm about the design CG.

PERFORMANCE AND SPEED STAGES

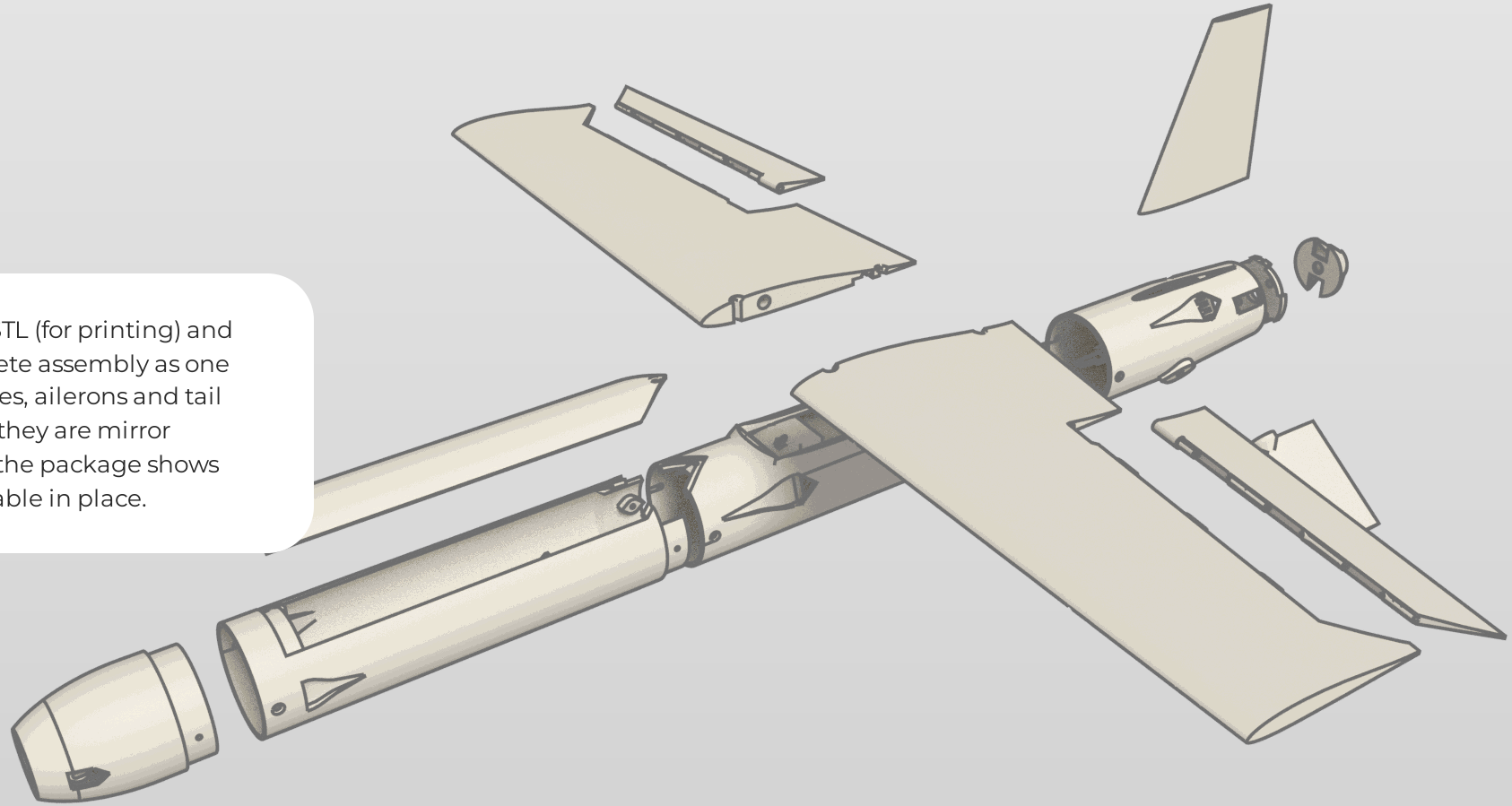
Motor	T-Motor F50 2207 2150KV
Propeller	Gemfan 5x7.5E Vortex
Battery	BetaFPV LAVA II 3S 680 95C LiHV
Static thrust	1458 g at 60 A (thrust / weight 3.66)
rpm governor / current limit	38,000 rpm max / 60 A
Top speed	118 m/s as printed, 120 m/s finished
Full-throttle time	about 0.6 min per pack pair (passes of ≤ 8 s)

Stage Vne	GOV_RPM (full stick)	Expected level speed
30 m/s	10,400 rpm	30 m/s
35 m/s	12,000 rpm	35 m/s
40 m/s	13,500 rpm	40 m/s
45 m/s	15,000 rpm	44 m/s
50 m/s	16,600 rpm	50 m/s
55 m/s	18,100 rpm	54 m/s
60 m/s	19,700 rpm	60 m/s
65 m/s	21,200 rpm	64 m/s
70 m/s	22,700 rpm	69 m/s
75 m/s	24,300 rpm	74 m/s
80 m/s	25,800 rpm	79 m/s
85 m/s	27,300 rpm	84 m/s
90 m/s	28,800 rpm	89 m/s
95 m/s	30,300 rpm	94 m/s
100 m/s	31,900 rpm	99 m/s
105 m/s	33,400 rpm	104 m/s
110 m/s	34,900 rpm	109 m/s
115 m/s	36,400 rpm	114 m/s
120 m/s	37,900 rpm	118 m/s

No airspeed sensor: each stage is a GOV_RPM value flown at full stick (switch on STAGE). Level runs; GPS two-way passes.

FILES

Every printed part is supplied as STL (for printing) and STEP (for editing), plus the complete assembly as one STEP file. Left and right wing halves, ailerons and tail panels are separate files because they are mirror images. The interactive viewer in the package shows every part, purchased item and cable in place.



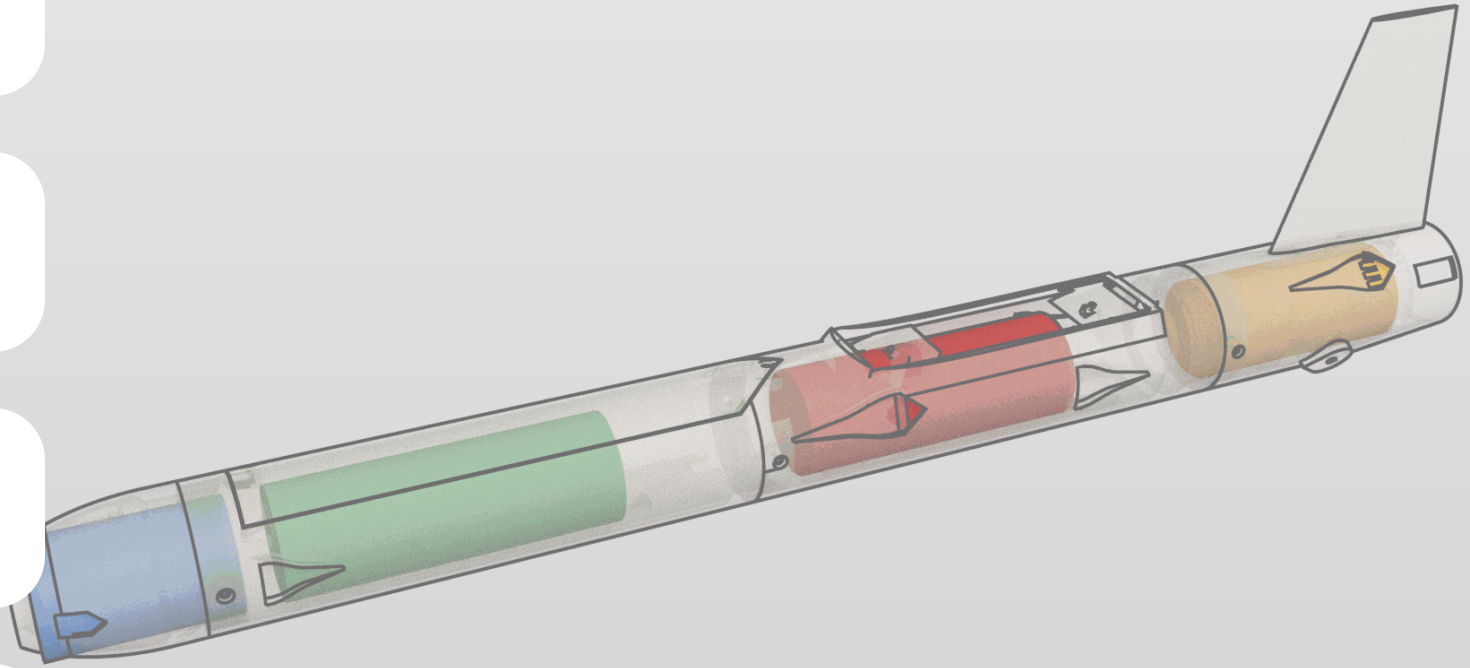
COMPARTMENT LAYOUT

BATTERY BAY
6S as two 3S packs side by side, 81 × 31 × 20 mm together; slides x 108–120 for balance; rear stop at x 160

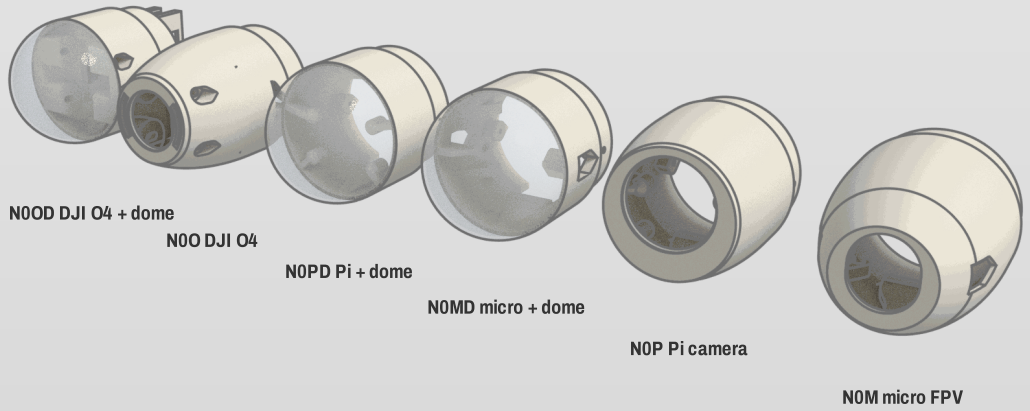
WING BAY
Flight controller stack with the ESC on top, receiver, aileron servo; reached with the wing off

TAIL MODULE
Two tail servos, pushrods, horns and the motor; plugs off at the MR30 and servo plugs

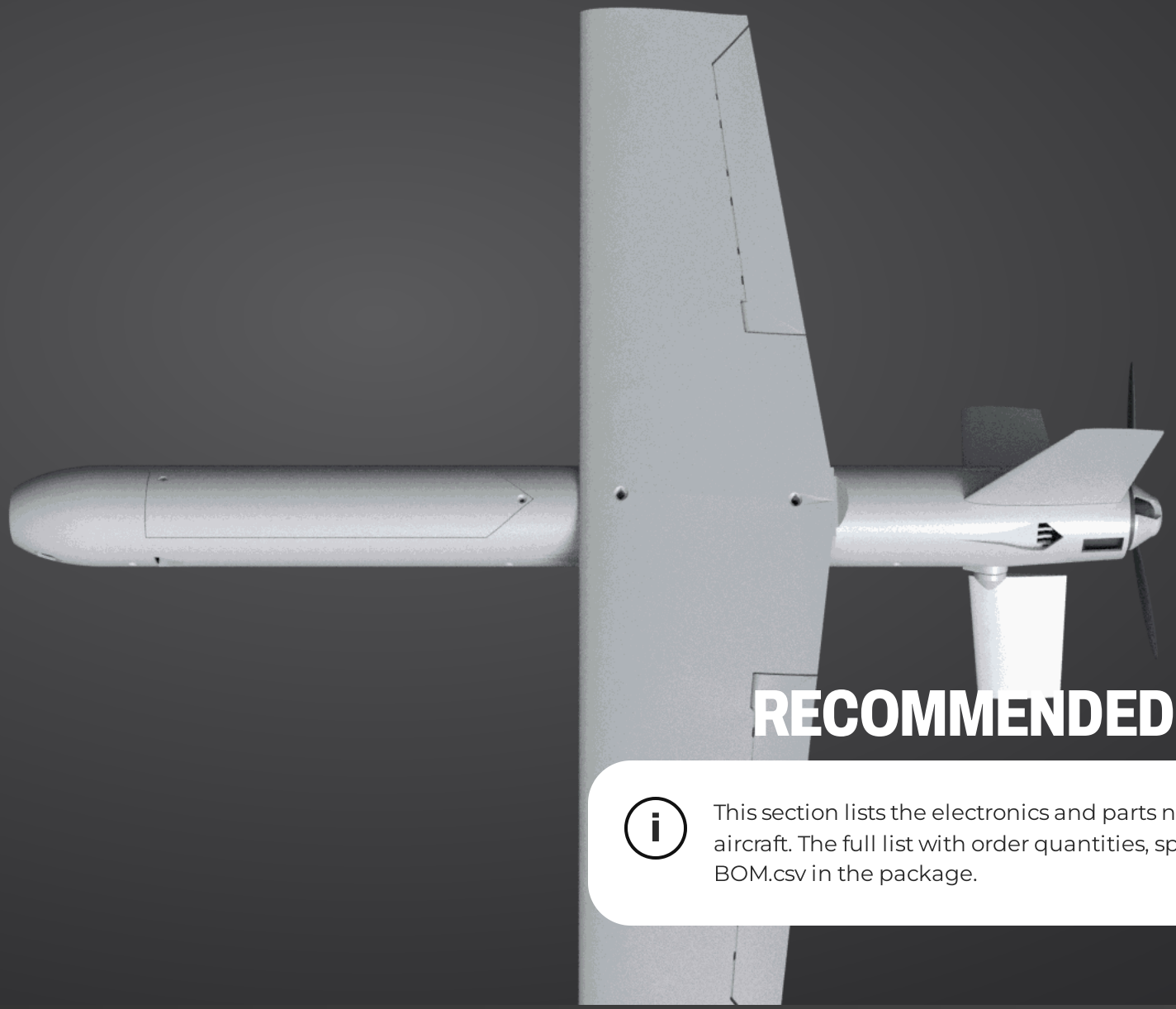
NOSE BAY
Camera and VTX (or the DJI O4 unit) in the removable nose



CAMERA NOSE VARIANTS



Nose	Camera	Battery centre for balance
N0M	19 mm micro FPV camera + 20 × 20 VTX	x 120 mm
N0P	Raspberry Pi Camera Module 3 (test option)	x 108 mm (at its stop, no ballast)
N0MD	micro camera behind a 40 mm dome, nano VTX under the hatch	x 124 mm (at its stop + tail ballast)
N0PD	Pi camera behind a dome	x 113 mm
N0O	DJI O4 Air Unit (Lite), ducted	x 125 mm (at its stop + tail ballast)
N0OD	DJI O4 behind a dome	x 128 mm (at its stop + tail ballast)



RECOMMENDED EQUIPMENT

This section lists the electronics and parts needed to complete the aircraft. The full list with order quantities, specifications and sources is BOM.csv in the package.

RECOMMENDED ELECTRONICS

COMPONENT	MODEL / SPECIFICATION
Motor	T-Motor F50 2207 2150KV
Propeller (fixed)	Gemfan 5x7.5E Vortex
ESC	Sequre SQESC 2670 70A (AM32)
Flight controller	Matek H743-WLITE
GPS + compass	TBS M10Q GPS GLONASS (QMC5883 compass), 13.5 g
Servos	3 × Emax ES9052MD or ES9052MD HV (Promodelhobby, Turkey)
Flight battery	2 × BetaFPV LAVA II 3S 680 95C LiHV (2-pack)
Receiver	RadioMaster RP1 V2
FPV camera	Micro 19 mm FPV camera, e.g. RunCam Phoenix 2 or Caddx Ratel 2
Video transmitter + antenna	20x20 analog or digital VTX with U.FL antenna
DJI O4 Air Unit (Lite)	DJI O4 Air Unit

RECOMMENDED ACCESSORIES

ITEM	QTY
XT30 connector pairs	2
Carbon tube	1
Music wire 3/32 in (2.38 mm)	2
Pushrods 1.2 mm with Z-bend	4
Pushrod keepers for 1.2 mm wire	4
Z-bend pliers, 1.1 mm drill, 1.2 mm drill (as a hand reamer), small flat file	0
Nylon countersunk screw M3 x 16	1
M3 hex nut	2
M2 x 6 countersunk screws (joints)	6
M2 x 6 countersunk screws (front hatch)	2
Kapton / thin clear tape	as needed
Adhesives	as needed
Velcro strap + hook/loop	1
Heat shrink + silicone wire	as needed

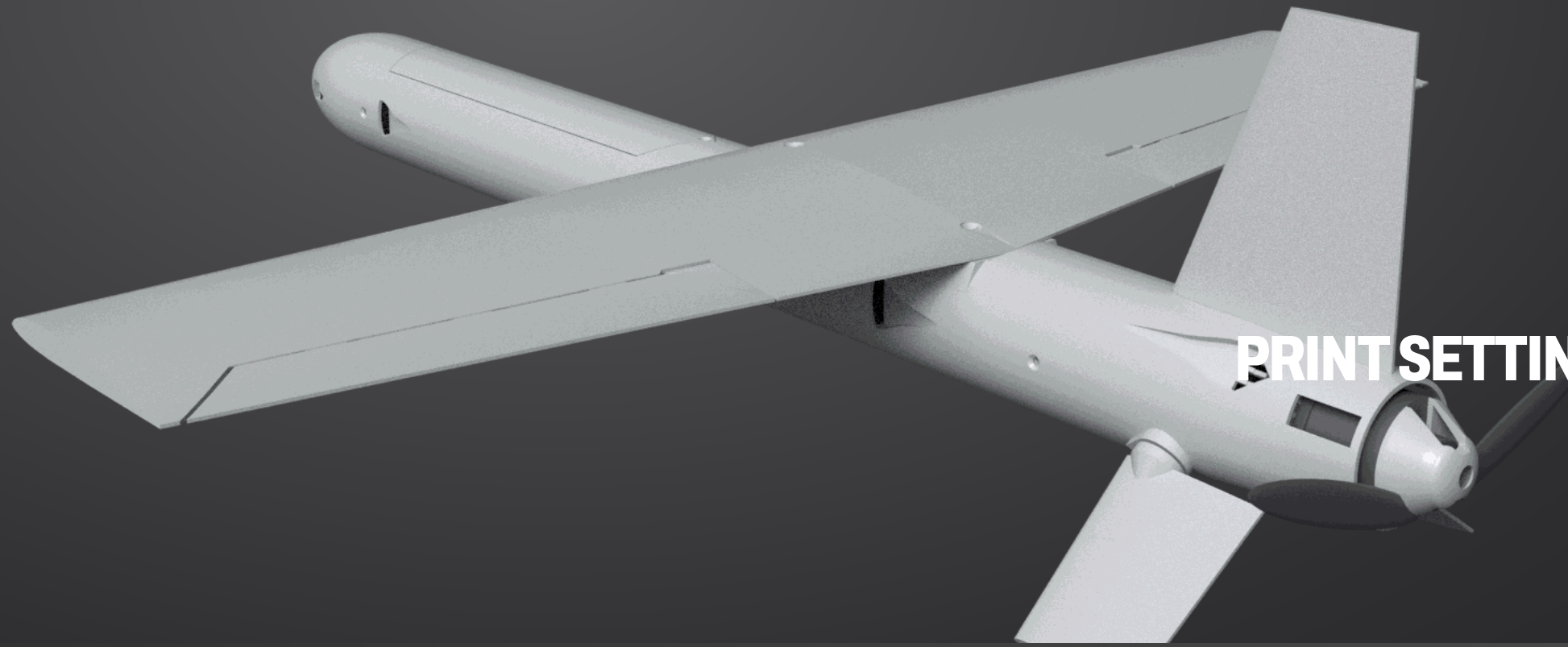
ITEM	QTY
Silver steel rod 2.5 mm	2
Brass tube 3 mm	2
MR30 connector pairs	1
Servo extension leads 100 mm	2
Motor screws M3	4
Glass skin + finish	25 g
Nylon countersunk screw M3 x 8	1
M2 x 5 screws	4
Servo strap / zip tie + foam tape	1
Steel rod 1.5 mm	2
Vent tape	1
15T servo arms	1
XT30 series adapter (2 into 1)	1

POWERTRAIN

Motor and prop. T-Motor F50 2207 2150KV (6S-rated, 24 V; Ø27 x 29.5 mm incl. shaft, 28.56 g, 16 x 16 mm M3 with 4 mm thread depth, 5 mm shaft, M5 x 10.75 mm prop thread, 50 mOhm, I0 1.6 A @10 V; 54.8 A / 1273 W peak): the SkyG Speed 60 motor, same mount. Prop: Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre nylon, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring; Robocombo stock). Gemfan publishes no rpm rating: each batch is spin-tested to 45,600 rpm (1.2 x the 38,000 rpm governor ceiling) behind a shield before flight. Pusher: the 5x7.5R, or the E mounted front face toward the motor with the motor reversed in the ESC; rpm-governed.

rpm governor and current limit. The flight controller runs skyg_governor.lua: it holds the prop at or below the stage rpm (never above 38,000 rpm, 92 % of the prop's rating) and the battery current at 60 A, the ESC and motor rating. In level flight the motor reaches about 37,862 rpm at full throttle; the governor acts in dives, at low speed on a full pack and per speed stage. THR_MAX 70 % is the backstop if the script stops. Prove it on the thrust stand first, and inspect the blades after every thrust-stand run and flying day.

Battery. BetaFPV LAVA II 3S 680 95C LiHV (2-pack); XT30; the two packs plug into a 2-into-1 XT30 series adapter. Full throttle in bursts of ≤ 10 s, then ≥ 30 s at low throttle: the ESC lives on its thermal mass. Static thrust in the model: 1458 g at 60 A; confirm on a thrust stand before the first flight.



PRINT SETTINGS

PRINT SETTINGS

The airframe is designed for **Bambu ASA Aero** (foaming, lightweight, heat deflection 85 °C) on a **0.4 mm nozzle** and **0.20 mm layers**; every part fits a Bambu H2S or H2D build volume. The motor plate and spinner are PETG-CF. ASA needs the closed chamber (45–60 °C), a dried spool (80 °C, 8 h) and glue on the plate; print long thin parts (wing halves, ailerons) with a brim against warping.

GENERAL GUIDELINES

- Fuselage, nose and hatch shells: 2 walls, no infill (the internal ribs and bosses are modelled).
- Wing halves and ailerons: 1 wall, 5 % gyroid.
- Tail panels: 2 walls, 10 % gyroid. Fin: 2 walls, 15 %.
- Motor plate and spinner: PETG-CF, 4 walls, 40 %.
- Tall standing parts: 5 mm brim, slow outer wall, enclosure door closed.

SUPPORTS: OFF

Every part is designed to print without supports in the orientation shown (DFAM rules, print audit: all parts pass). Keep supports switched off. The only bridge longer than 12 mm is under the rear wing-screw nut in F2, hidden inside the wing bay: if it sags on your printer, paint a small support under it there. Check the top of the aileron's torque-rod bore on the first aileron.

PRINT SETTINGS

FIRST PART AND WEIGHT CHECK
Print F1 first and weigh it. If it is more than 15 % over the sliced mass (about 12 g), lower the ASA Aero temperature or flow before printing the rest; foaming filaments vary with temperature, humidity and the enclosure.

COUPON TESTS
Before the full build, print the last 25 mm of F1 with the first 25 mm of F2, and the F2/F3 and nose joints the same way: every joint must slide on and off by hand and the M2 screws must bite. Print a 40 mm wing-root section and check the spar fit.

SLICER
The masses in this manual come from Bambu Studio with the H2S 0.4 mm profile and the walls/infill listed on the orientation pages. Import the STL files in millimetres; keep each part in the orientation shown.

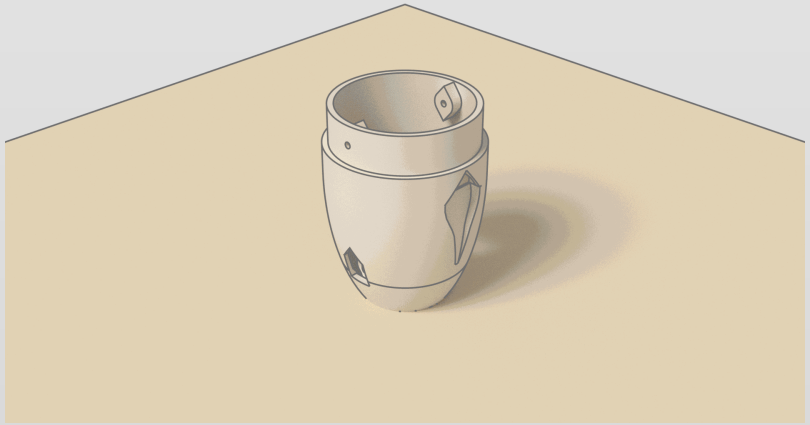
PARTS LIST

PART	MATERIAL	MASS
N0M NOSE (MICRO FPV)	ASA Aero	4.3 g
N0P NOSE (PI CAMERA)	ASA Aero	4.0 g
N0MD NOSE (MICRO + DOME)	ASA Aero	2.7 g
N0PD NOSE (PI + DOME)	ASA Aero	2.9 g
N0O NOSE (DJI O4)	ASA Aero	5.3 g
N0OD NOSE (DJI O4 + DOME)	ASA Aero	4.3 g
F1 FUSELAGE FRONT	ASA Aero	12.3 g
F2 FUSELAGE MAIN	ASA Aero	14.6 g
F3 TAIL MODULE	ASA Aero	7.2 g
HF FRONT HATCH	ASA Aero	2.7 g
FN FIN	ASA Aero	3.7 g
VTX SHELF (N0M)	ASA Aero	0.3 g
DOME RING	ASA Aero	0.3 g
HG GPS CARRIER	ASA Aero	0.8 g

PART	MATERIAL	MASS
WR WING HALF RIGHT	ASA Aero	10.9 g
WL WING HALF LEFT	ASA Aero	11.5 g
AR AILERON RIGHT	ASA Aero	1.5 g
AL AILERON LEFT	ASA Aero	1.3 g
SR TAIL PANEL RIGHT	ASA Aero	2.2 g
SL TAIL PANEL LEFT	ASA Aero	2.1 g
M1 MOTOR PLATE	PETG-CF	1.9 g
SP SPINNER	PETG-CF	1.9 g

One aircraft with the N0M nose: 79 g printed. Only one camera nose is needed; the others are options.

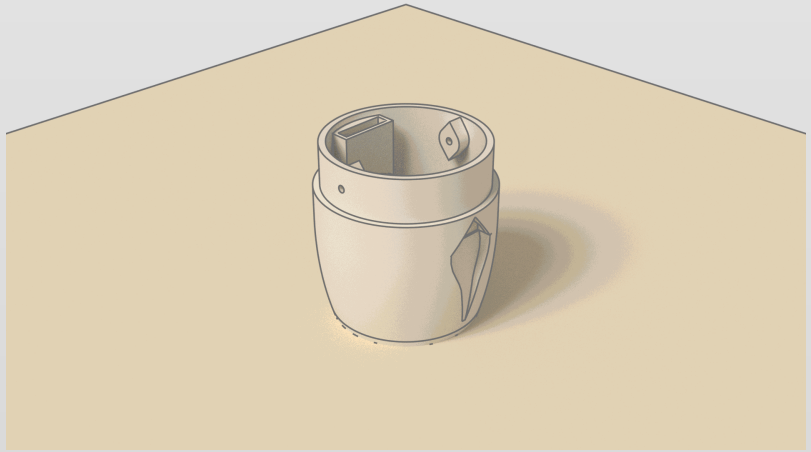
PARTS ORIENTATION



N0M NOSE (MICRO FPV)

FACE DOWN, SPIGOT UP

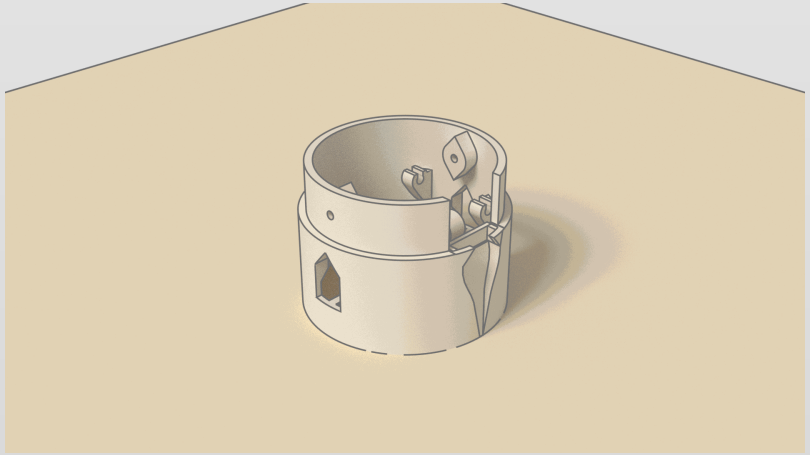
2 WALLS · 0 % INFILL · ASA AERO



N0P NOSE (PI CAMERA)

FACE DOWN, SPIGOT UP

2 WALLS · 0 % INFILL · ASA AERO

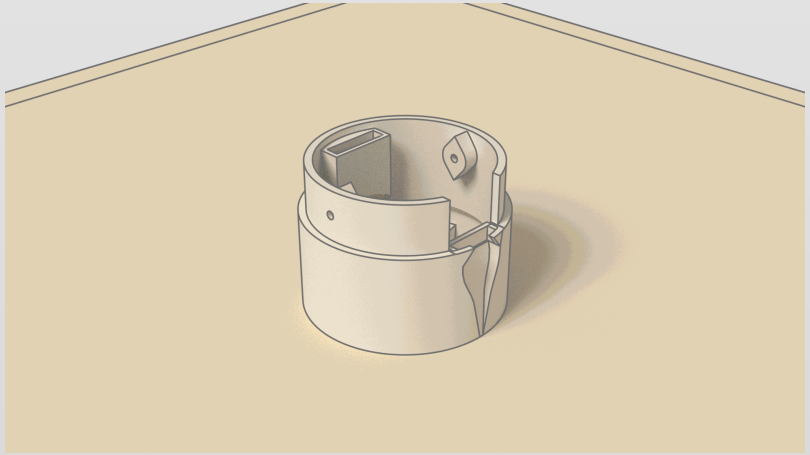


N0MD NOSE (MICRO + DOME)

FRONT FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

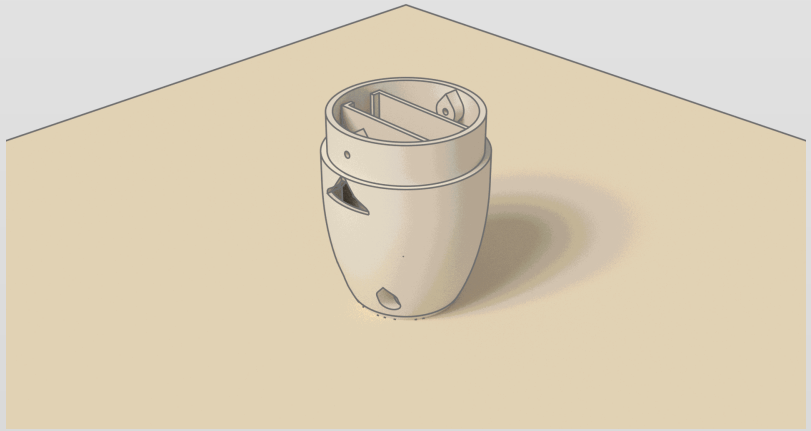
PARTS ORIENTATION



N0PD NOSE (PI + DOME)

FRONT FACE DOWN

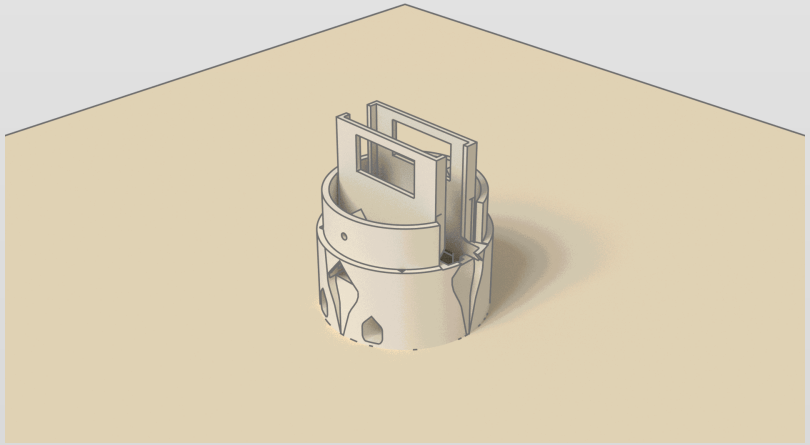
2 WALLS · 0 % INFILL · ASA AERO



N0O NOSE (DJI O4)

FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

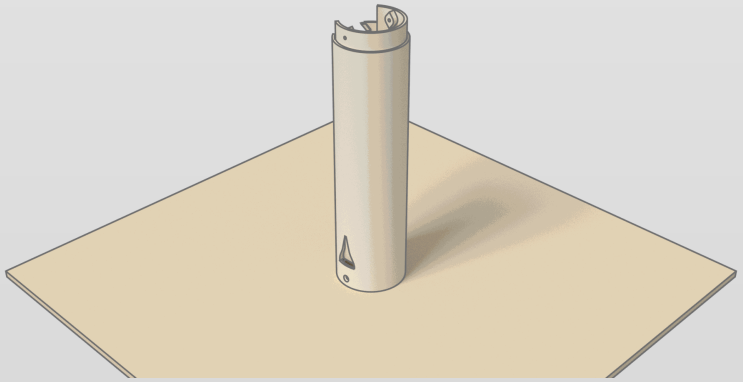


N0OD NOSE (DJI O4 + DOME)

FRONT FACE DOWN

2 WALLS · 0 % INFILL · ASA AERO

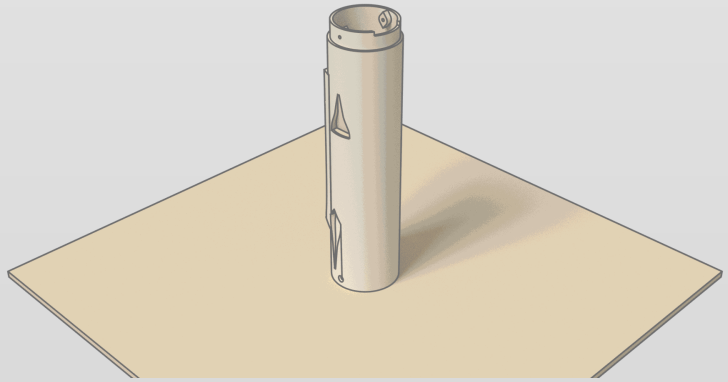
PARTS ORIENTATION



F1 FUSELAGE FRONT

FRONT END DOWN · SPIGOT UP · 5 mm BRIM

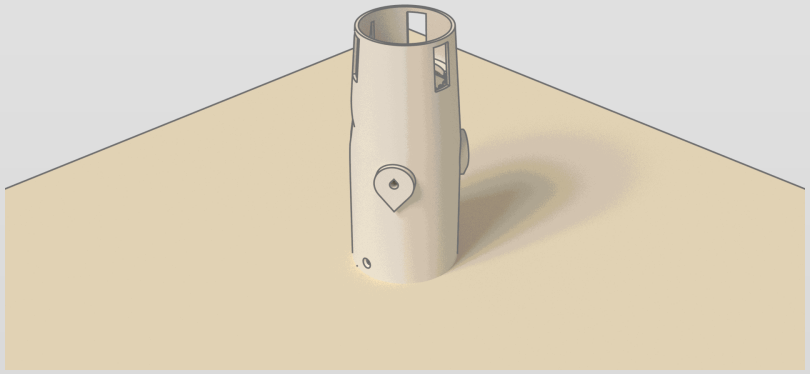
2 WALLS · 0 % INFILL · ASA AERO



F2 FUSELAGE MAIN

FRONT END DOWN · SPIGOT UP · 5 mm BRIM

2 WALLS · 0 % INFILL · ASA AERO

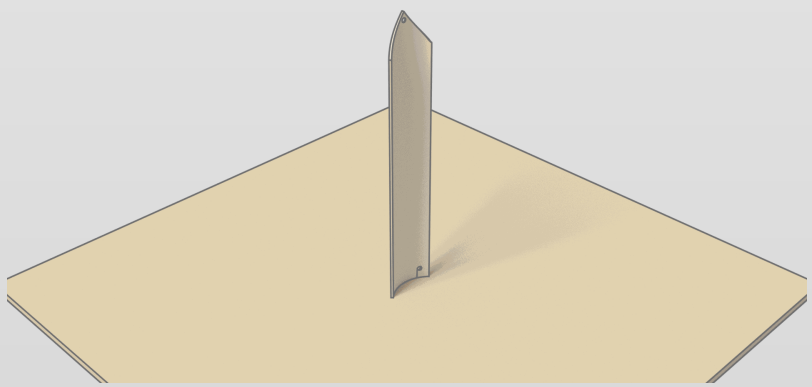


F3 TAIL MODULE

FRONT END (SOCKET) DOWN · 5 mm BRIM

2 WALLS · 0 % INFILL · ASA AERO

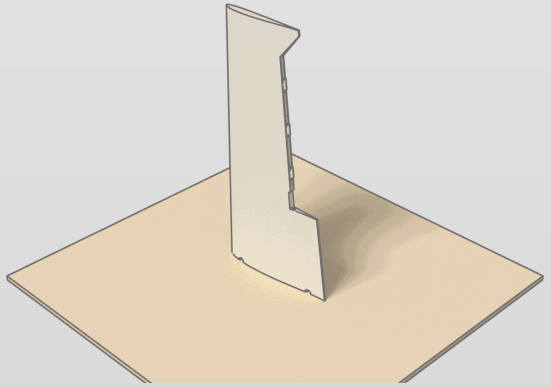
PARTS ORIENTATION



HF FRONT HATCH

STANDING, FRONT END DOWN

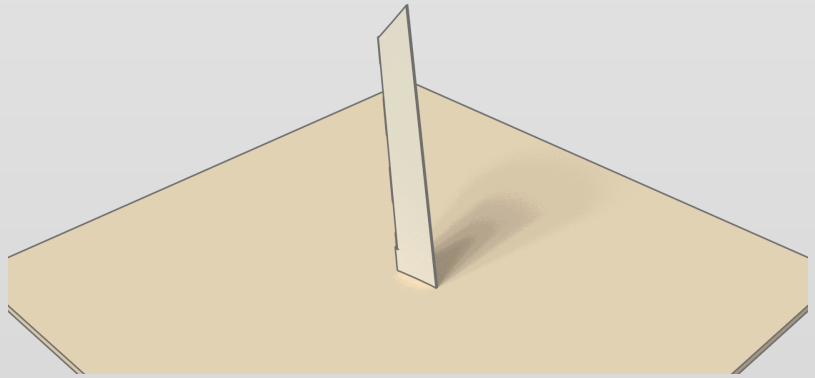
2 WALLS · 0 % INFILL · ASA AERO



WR WING HALF R / L

ON THE ROOT RIB

1 WALLS · 5 % INFILL · ASA AERO

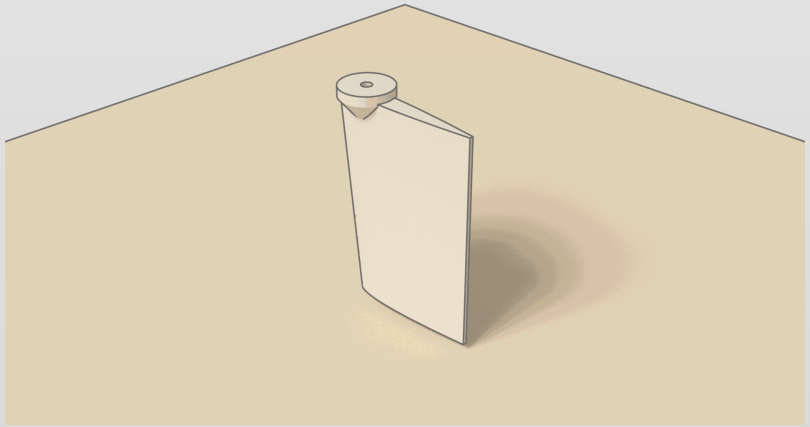


AR AILERON R / L

ON THE INBOARD END

1 WALLS · 5 % INFILL · ASA AERO

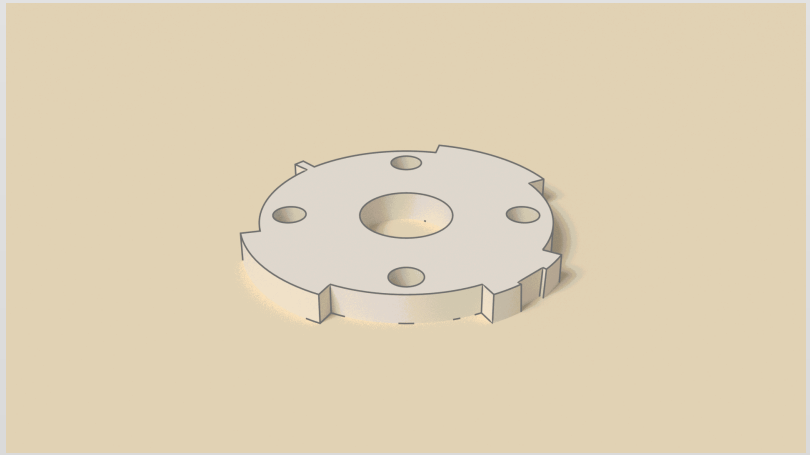
PARTS ORIENTATION



SR TAIL PANEL R / L

TIP DOWN, HUB UP

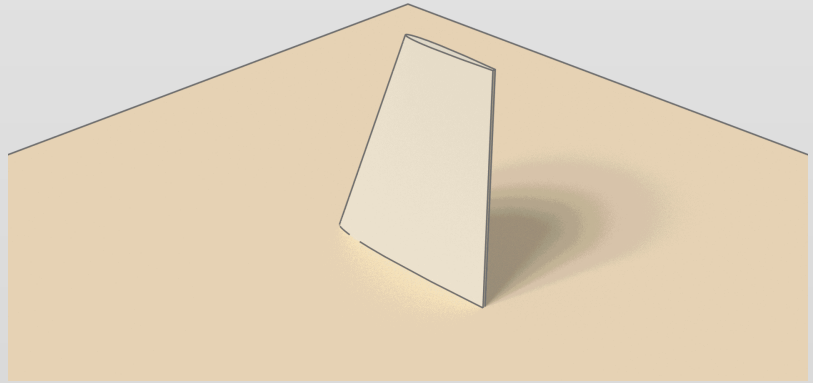
2 WALLS · 10 % INFILL · ASA AERO



M1 MOTOR PLATE

FLAT

4 WALLS · 40 % INFILL · PETG-CF

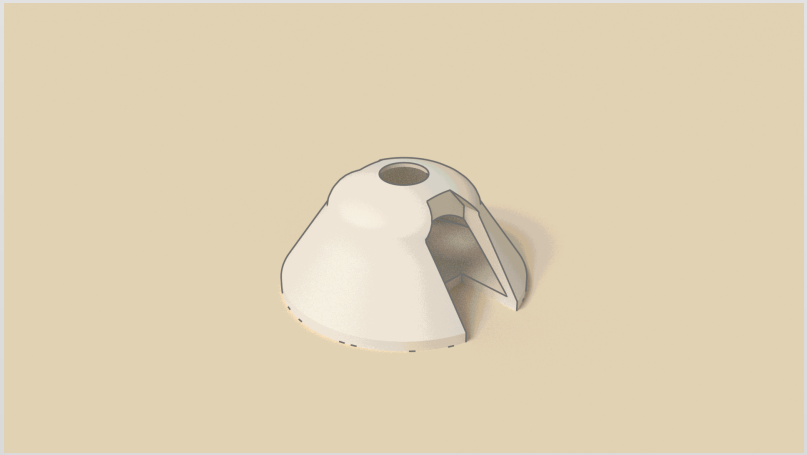


FN FIN

ON THE ROOT TAB

2 WALLS · 15 % INFILL · ASA AERO

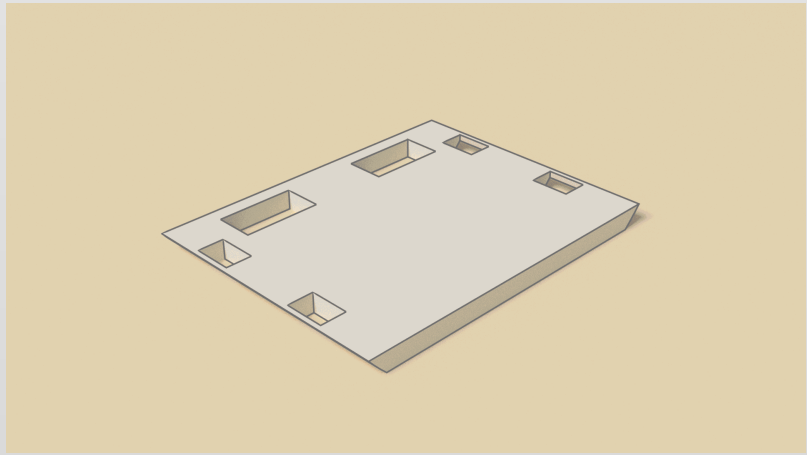
PARTS ORIENTATION



SP SPINNER

BACKPLATE DOWN

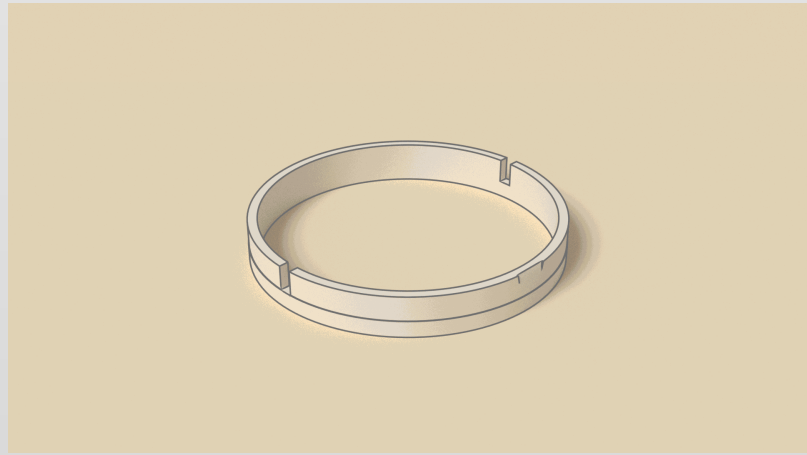
4 WALLS · 40 % INFILL · PETG-CF



VTX SHELF (N0M)

FLAT

2 WALLS · 15 % INFILL · ASA AERO

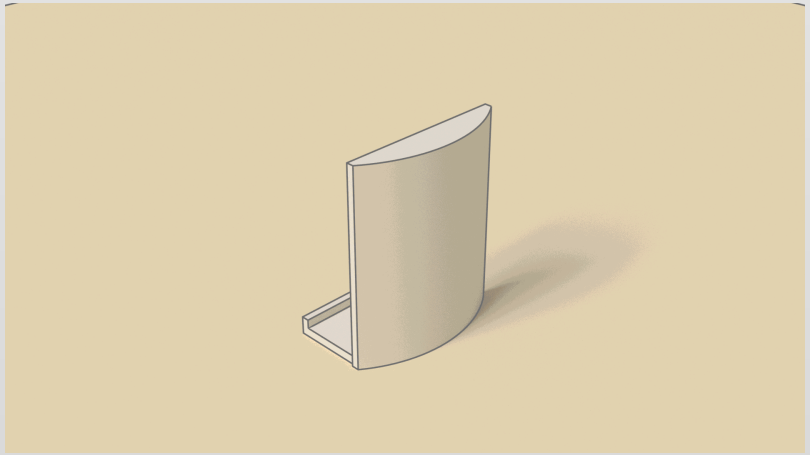


DOME RING

LIP DOWN

2 WALLS · 15 % INFILL · ASA AERO

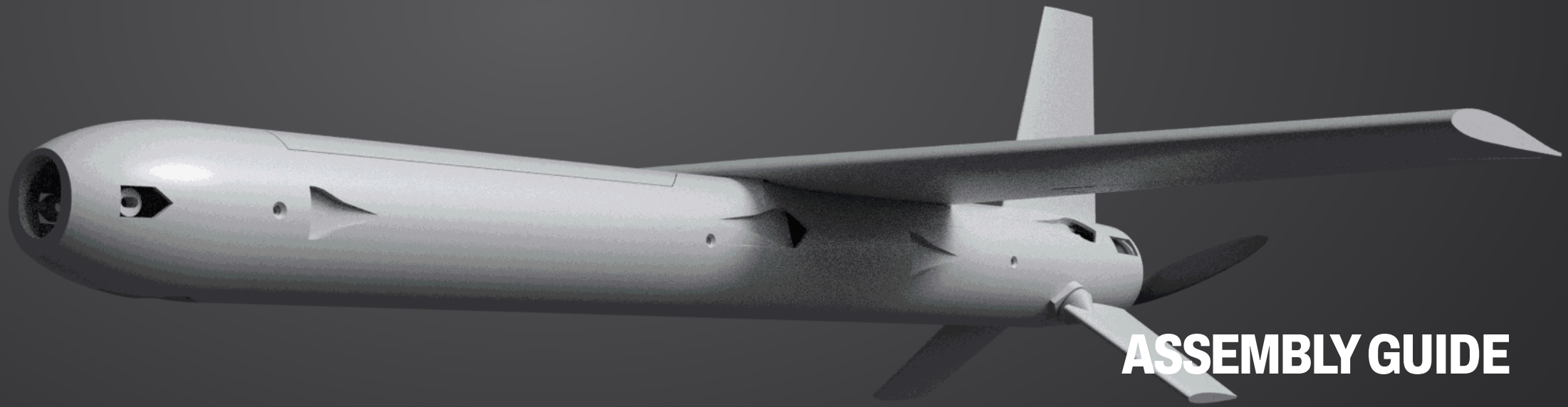
PARTS ORIENTATION



HG GPS CARRIER

ON ITS REAR WALL

2 WALLS · 15 % INFILL · ASA AERO

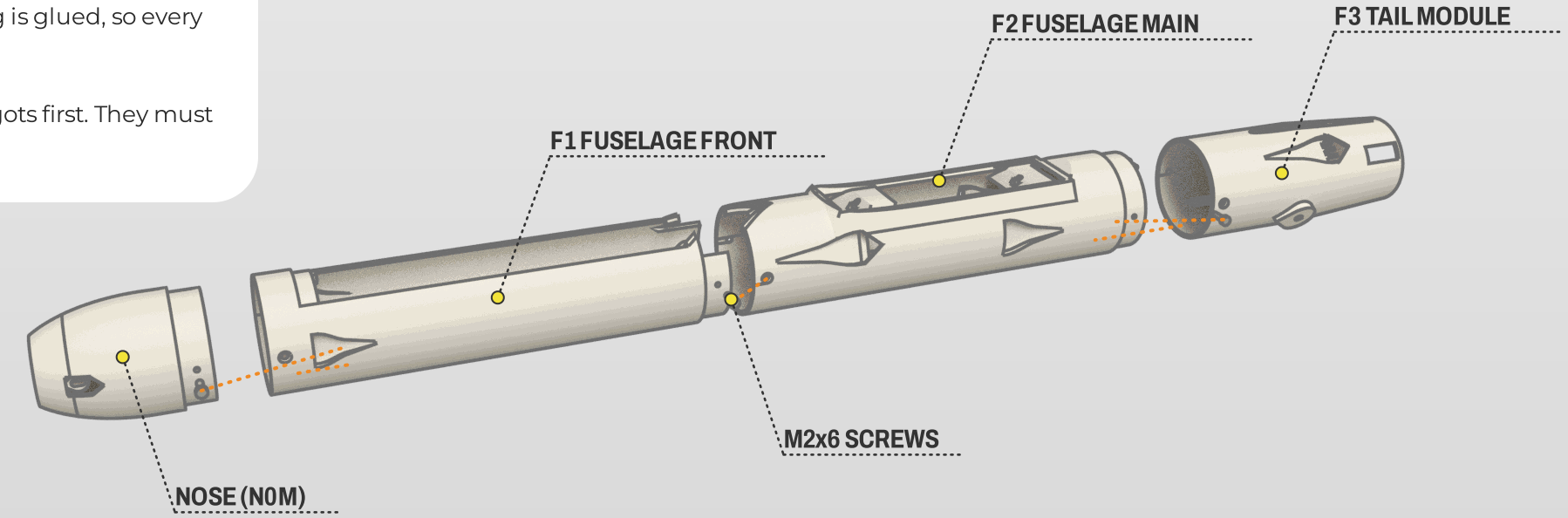


ASSEMBLY GUIDE

FUSELAGE ASSEMBLY

The fuselage is four modules on spigot joints: the camera nose, F1, F2 and the F3 tail module. Each joint is held by two M2 × 6 countersunk screws from the sides into bosses inside the spigot, their heads flush in the countersinks; nothing is glued, so every module comes off for service.

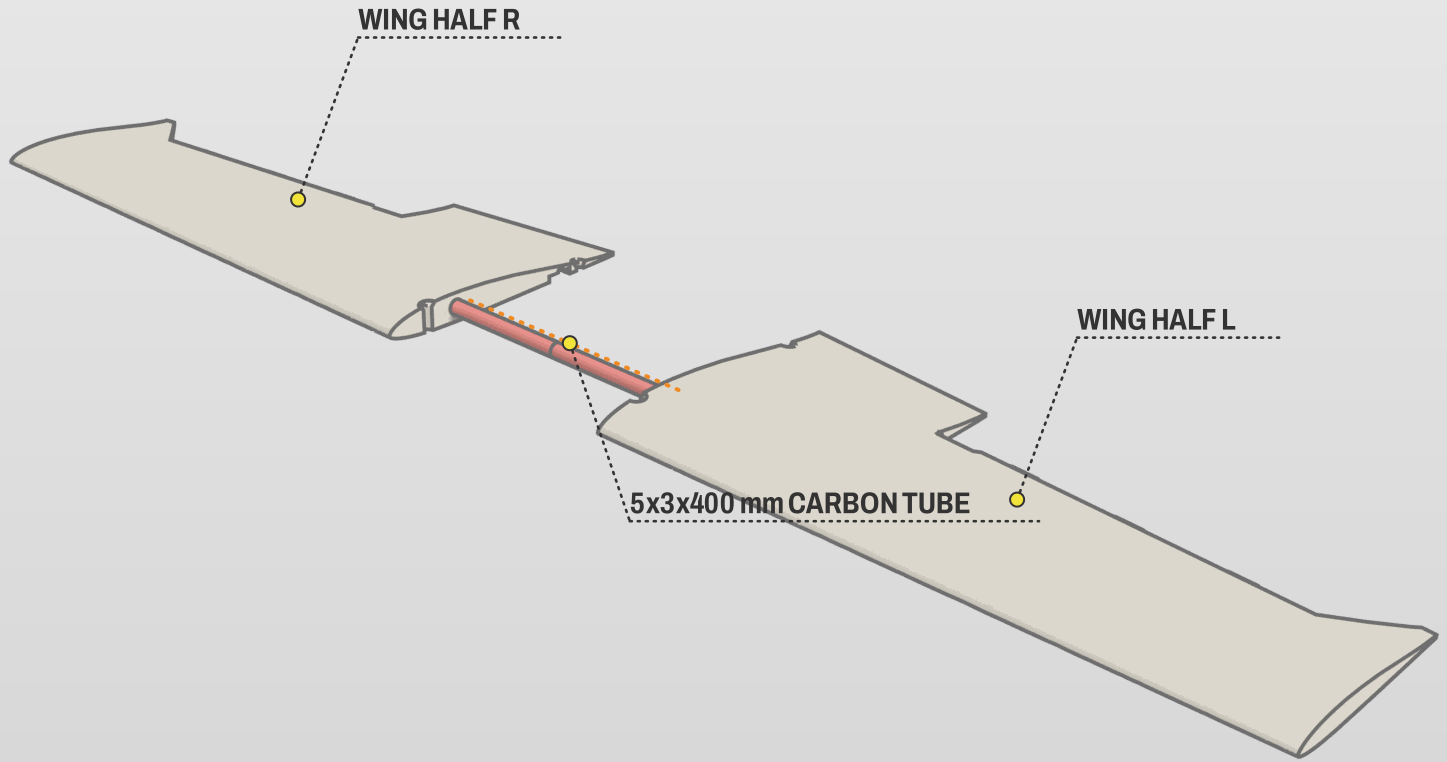
Clean any stringing from the spigots first. They must slide in by hand.



M2 × 6 SCREWS

WING ASSEMBLY

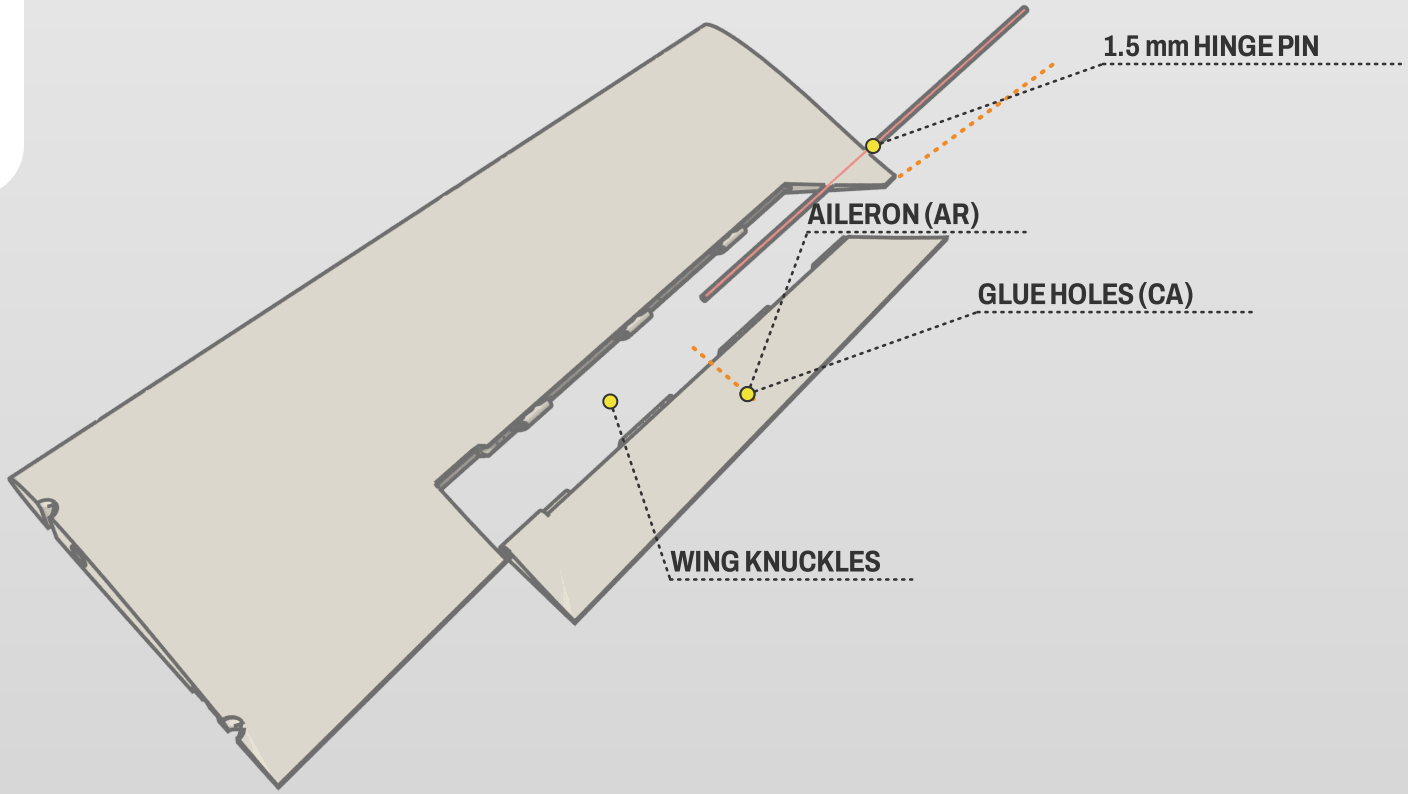
Cut one 400 mm length of 5 × 3 mm carbon tube and slide it through both wing halves (200 mm into each). Epoxy the tube in the bores and the two root ribs together with the whole wing flat on a table (0° dihedral).



EPOXY

WING ASSEMBLY

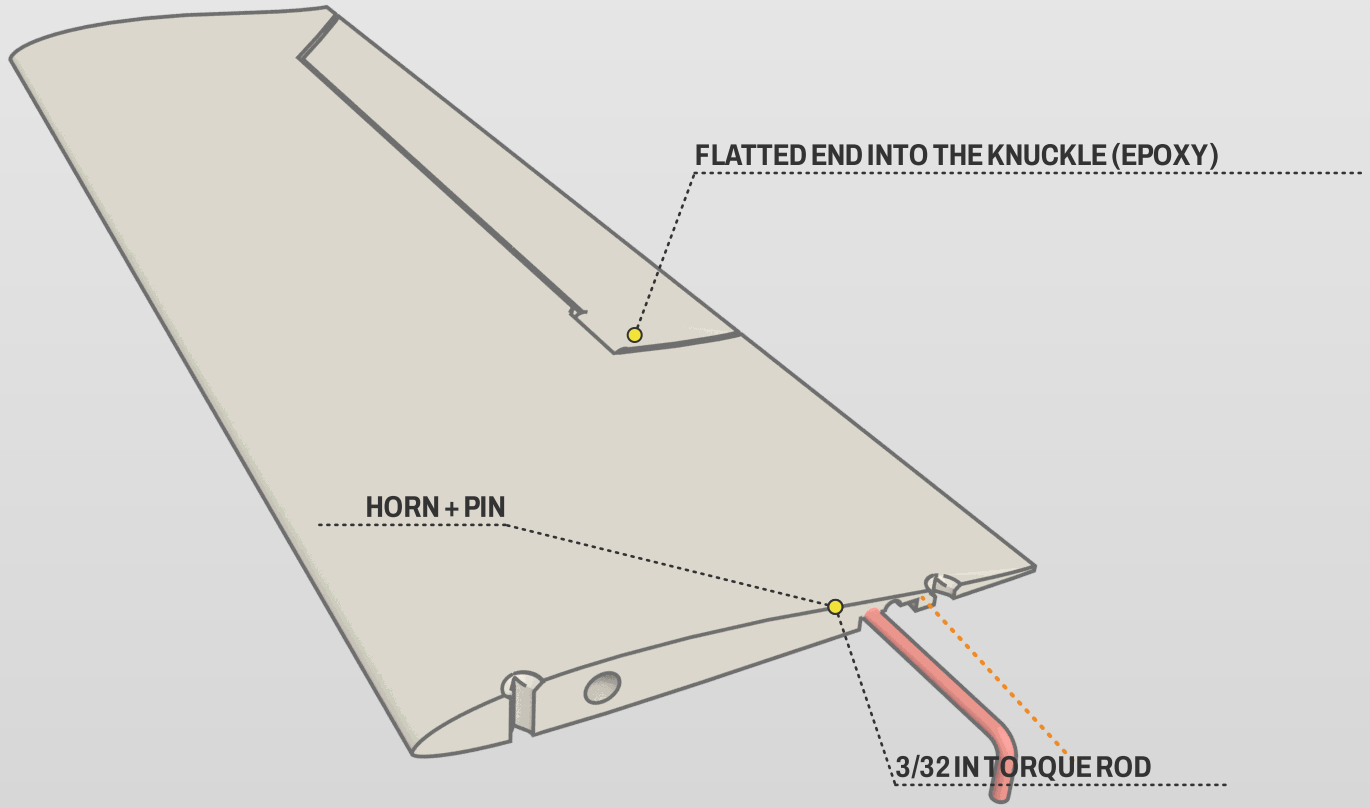
Set each aileron in place: its rounded nose sits in the cove along the wing's hinge face and the wing's three knuckles enter the pockets in the nose. Push a 132 mm length of 1.5 mm steel rod in from the wingtip through the tip, the aileron and the knuckles. Wick thin CA into the aileron's four glue holes only, then plug the tip hole. The aileron must swing freely.



THIN CA

WING ASSEMBLY

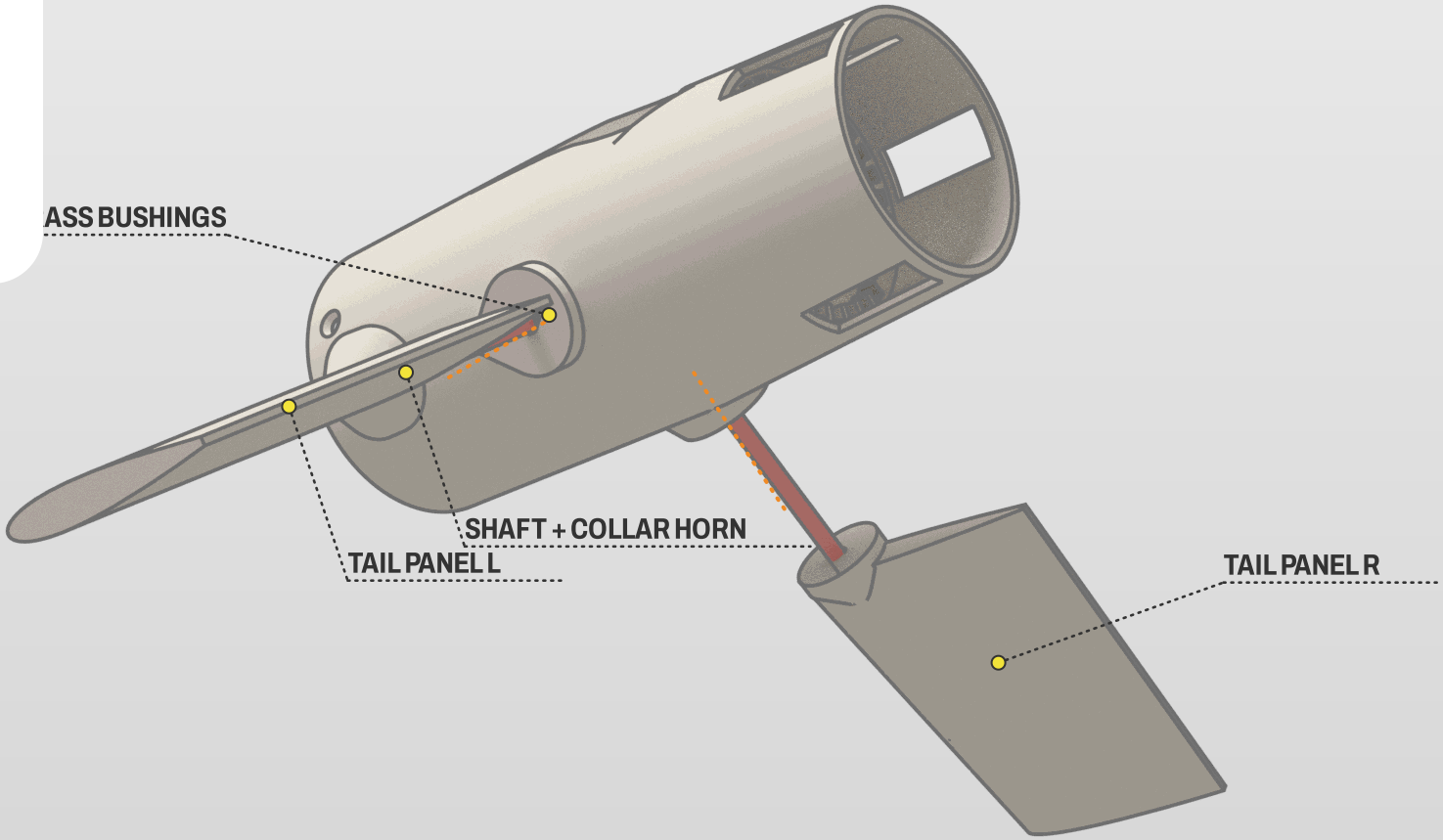
Bend a 12 mm horn down at one end of each 3/32 in (2.38 mm) rod with a 5.5 mm pin outboard at its tip, and file a 0.3 mm flat on the last 9 mm of the straight end, opposite the horn. Push it in from the root face until the flat enters the D-bore of the aileron's root knuckle; epoxy the end and hold the aileron 3° trailing edge up while it cures.



EPOXY

TAIL MODULE

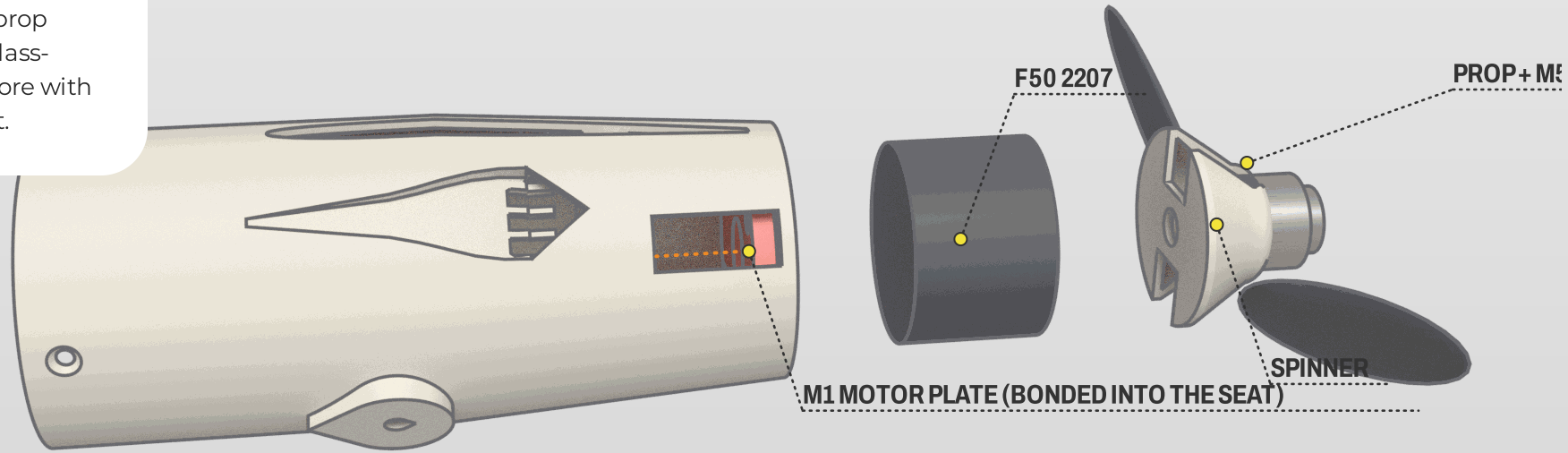
Press a 10 mm brass bushing into each side boss of F3. Epoxy a 2.5 mm steel shaft 35 mm into each tail panel's root bore. From the open front of F3, push each shaft through its bushing and fit the wheel-collar horn with its 8 mm brass plate standing along the panel's normal (not pointing forward), 0.5 mm end play, set screw on a filed flat.



EPOXY

TAIL MODULE

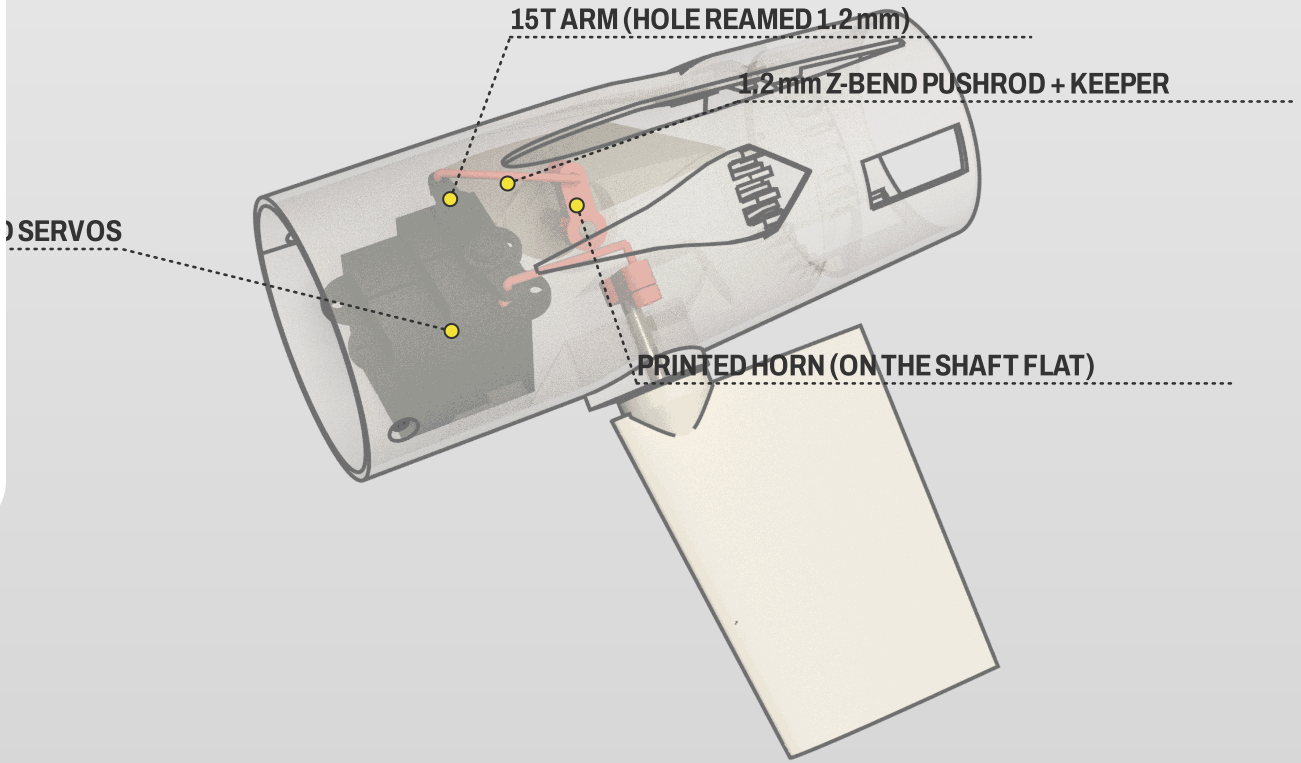
Bond the M1 plate into the conical seat at the rear of F3 with high-temperature epoxy, its edge notches lined up with the cooling windows; it is the motor bulkhead. Then screw the motor to it with its 4 × M3 from inside the module, through the 28 mm opening. Do this before the tail-panel shafts and servos go in. Fit the spinner backplate, the prop (Gemfan Vortex 5x7.5E fixed electric prop (glass-fibre nylon, 3.7 g, hub 13 x 6 mm, 6.35 mm bore with the 5 mm adapter ring) and the thin M5 nut.



EPOXY

TAIL MODULE

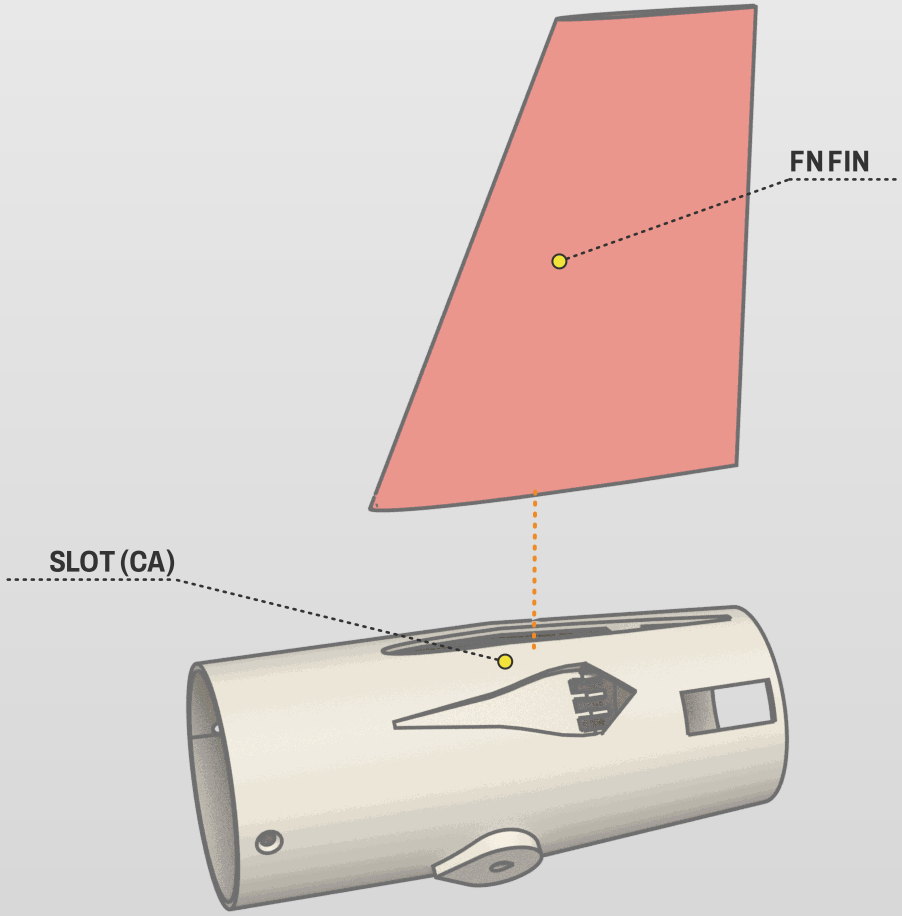
Stand both ES9052MD servos through the cut-outs in the F3 floor on foam tape on the bottom skin, a strap over them, output at the aft end, leads along the side walls. One-sided 15T arm pointing outboard; file a 0.3 mm flat on the inboard end of each 2.5 mm stabilator shaft and epoxy the printed horn (THR / THL) on it, arm along the panel's normal. Drill the servo arm's 6.7 mm hole 1.1 mm and ream it by hand with a 1.2 mm bit (and the horn's hole the same): the wire must turn with no play. 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend at the horn end pushed outboard through the horn's reamed hole, snap-on keeper on the protruding end (unclip it to take the tail module off). Free play at each panel trailing edge <= 0.2 mm. Check both panels move freely through ±20°.



SERVO TAPE

TAIL MODULE

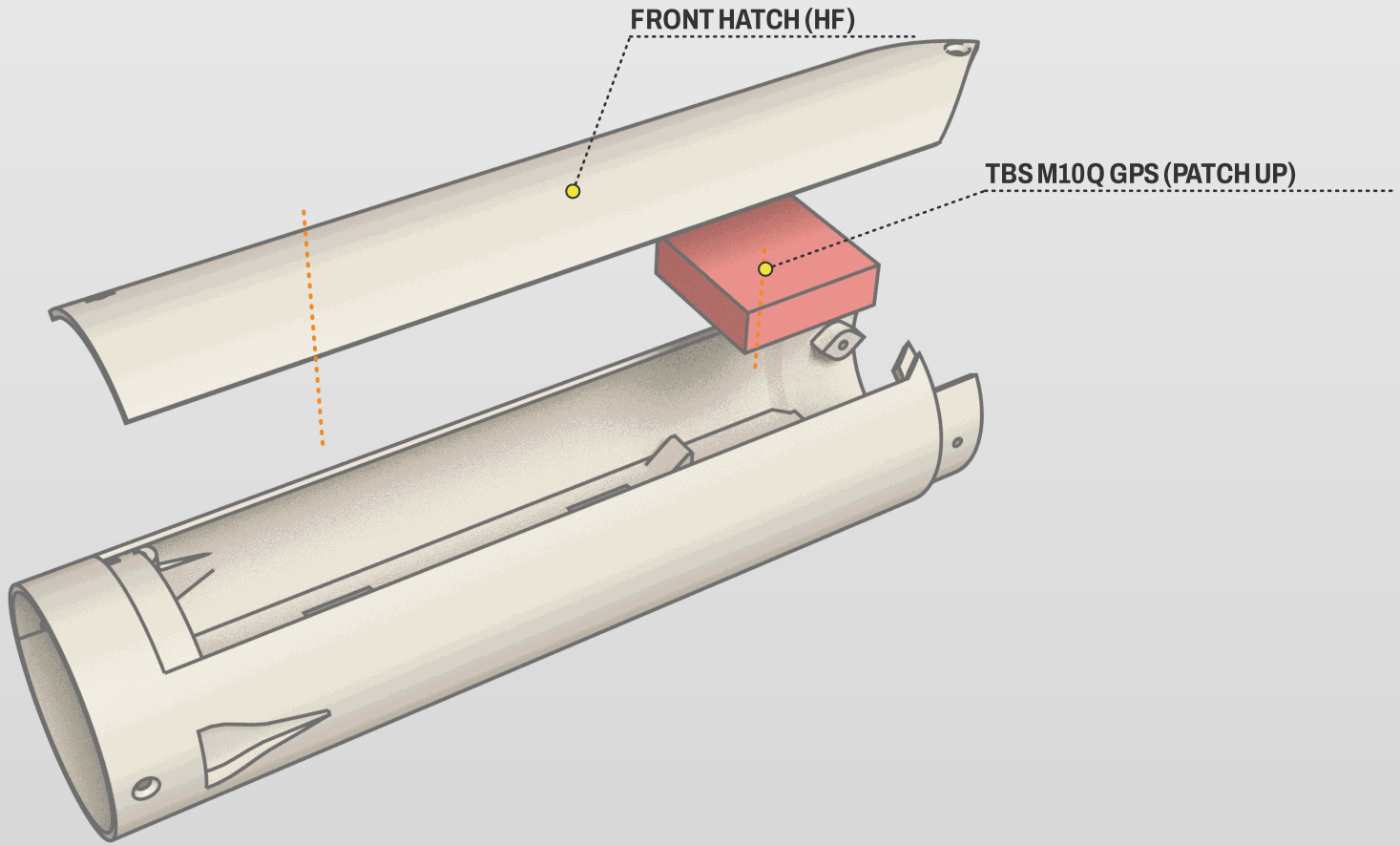
Glue the fin into the top slot of F3 with CA, vertical and square to the wing.



CA GLUE

FUSELAGE ASSEMBLY

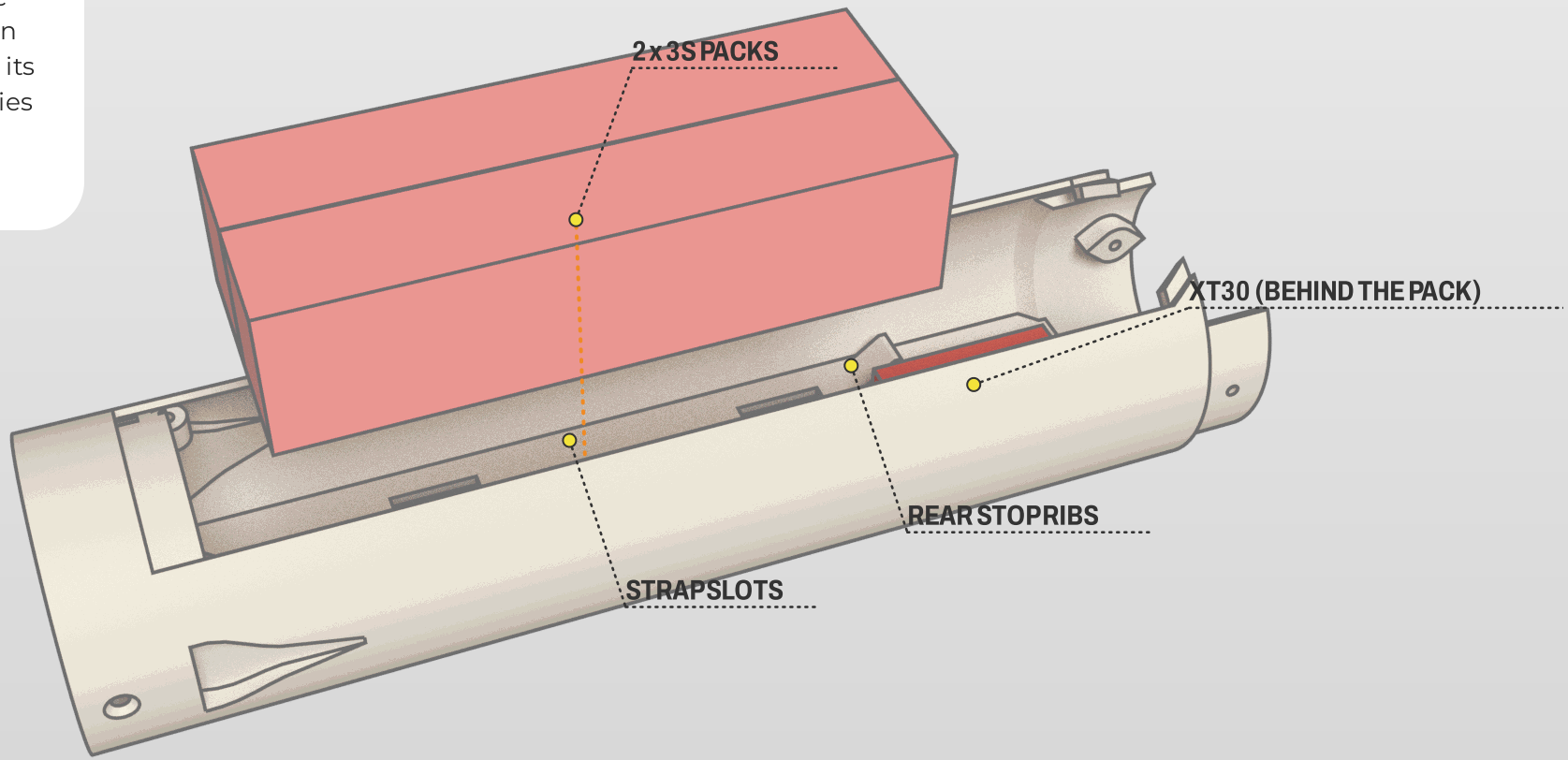
Bond the printed GPS carrier (HG) under the rear of the front hatch skin. Foam-tape the TBS M10Q GPS, patch up, to its ceiling, against its rear ledge. The hatch's pointed rear end reaches into F2 and its rear screw goes into F2: take the hatch off before separating F1 and F2. Close the hatch with the two M2 × 6 countersunk screws at its ends (they self-tap into the bosses under the lips; do not overtighten). Leave a slack loop in the GPS lead so the hatch lifts off.



M2 × 6 SCREWS

FUSELAGE ASSEMBLY

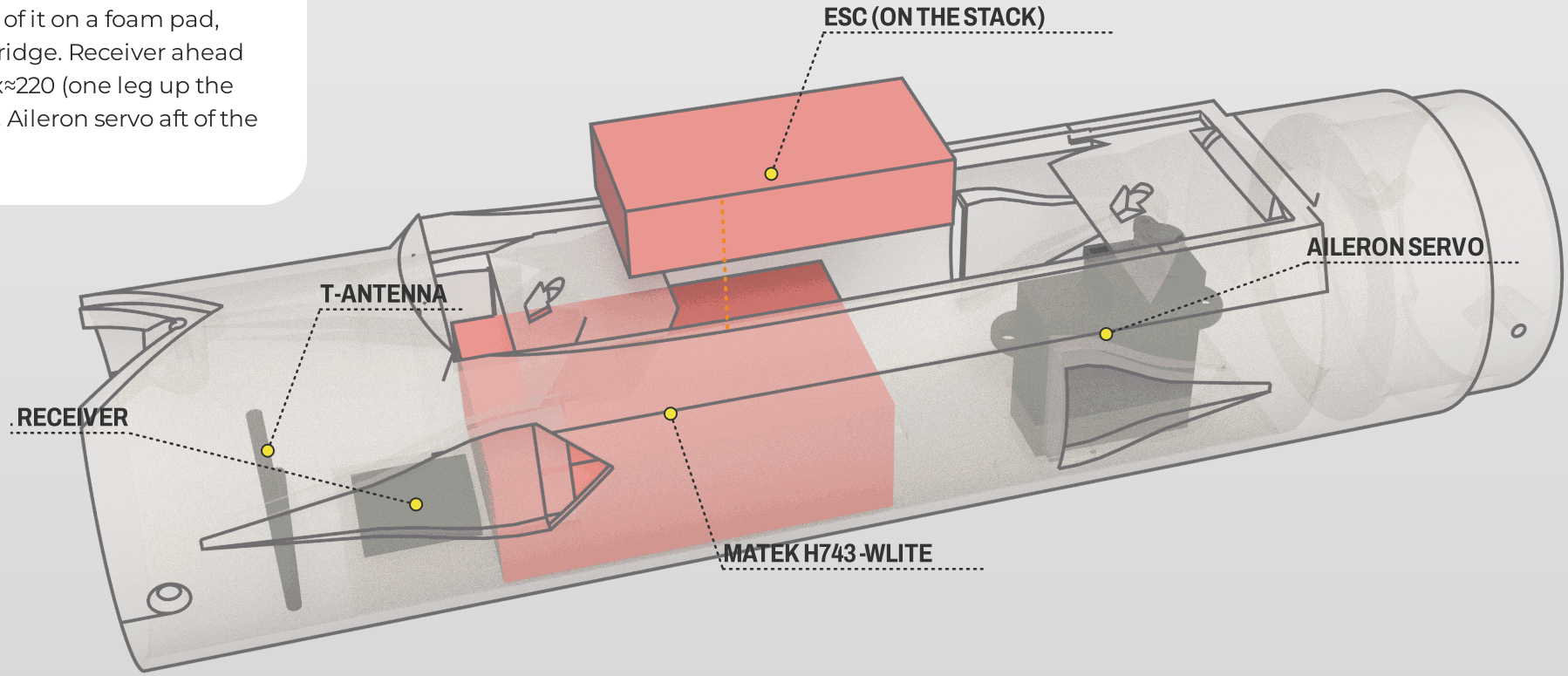
The two 3S packs sit side by side on their sides on the F1 floor, joined by the XT30 series adapter; one strap through the slots holds both. The pack sits on the F1 floor with a strap through the slots, leads at its rear. Two floor ribs are its rear stop; the XT30 pair lies behind the pack under the GPS and unplugs through the front hatch.



VELCRO STRAP

EQUIPMENT

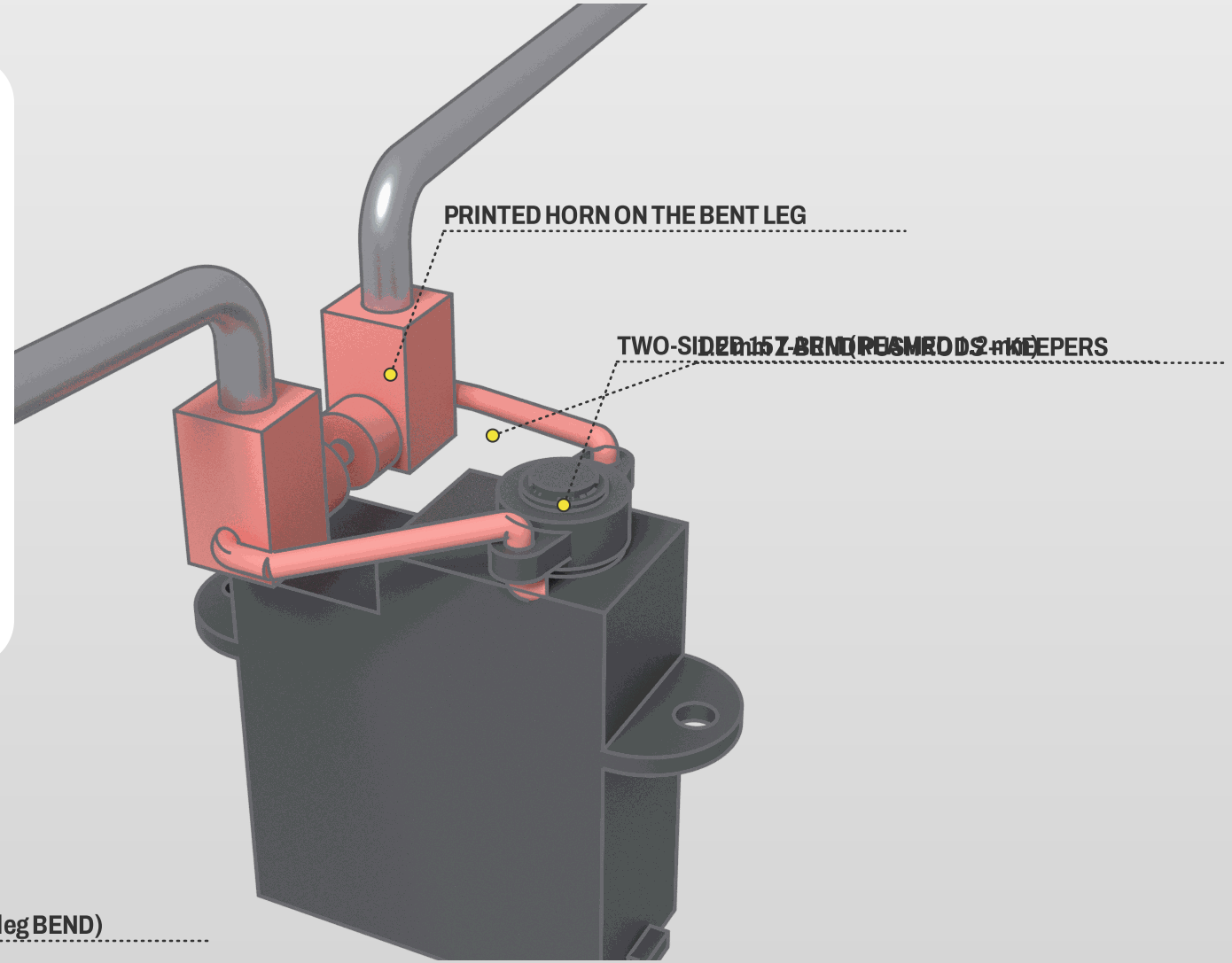
Matek H743-WLITE on a 0.5 mm foam pad and a strap on the wing-bay floor, arrow forward, servo rail (Vx) set to 6 V; the ESC on top of it on a foam pad, behind the front wing-bolt bridge. Receiver ahead of it, its T-antenna as an L at x≈220 (one leg up the left wall, one across the floor). Aileron servo aft of the stack, output at the aft end.



FOAM TAPE

EQUIPMENT

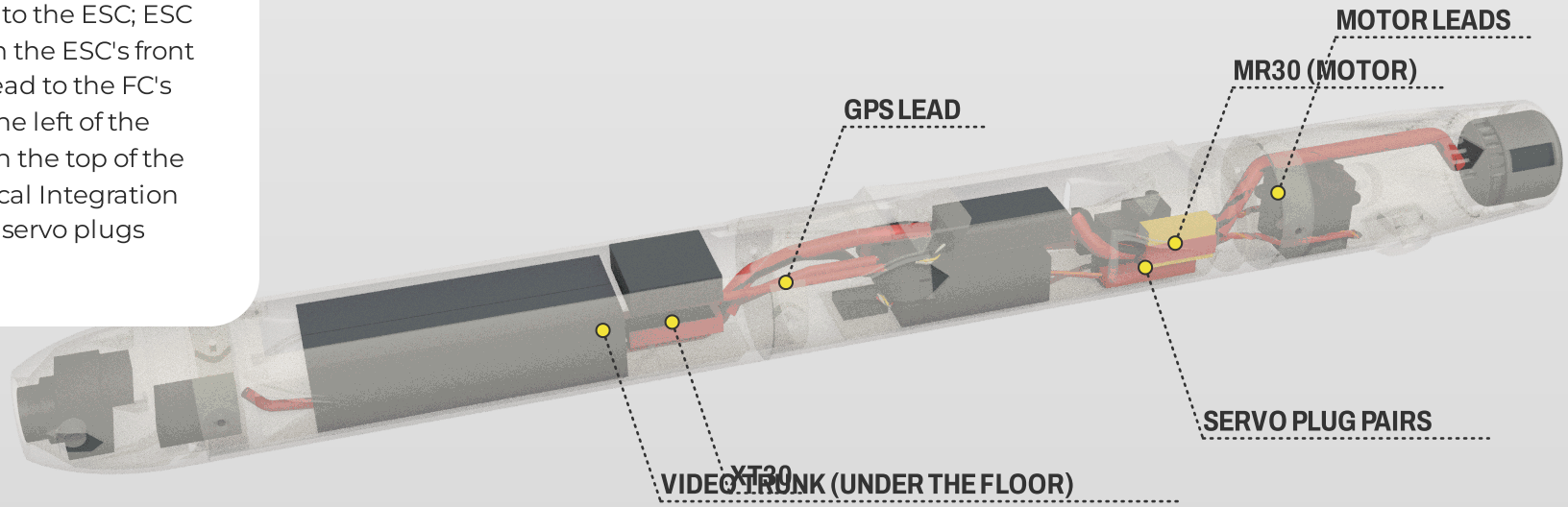
ES9052MD aileron servo standing through the wing-bay floor on foam tape; two-sided 15T arm. Each torque rod has one 90° bend (the horn leg, 6 mm from the centreline); epoxy the printed horn (AHR / AHL) on the leg, 4 mm deep. Drill the arm's 4.3 mm holes 1.1 mm and ream them by hand with a 1.2 mm bit (and both horns' holes): the wire must turn with no play. Each horn connects to the arm with a 1.2 mm Z-bend pushrod: Z-bend in the arm, 90° L-bend pushed inboard through the horn's reamed hole, snap-on keeper on the protruding end. Check: with the servo powered, each aileron trailing edge moves <= 0.2 mm (free play lowers aileron flutter to about 33 m/s). Set both ailerons 3° trailing edge up at servo neutral. No play at either end.



Z-BENDS

EQUIPMENT

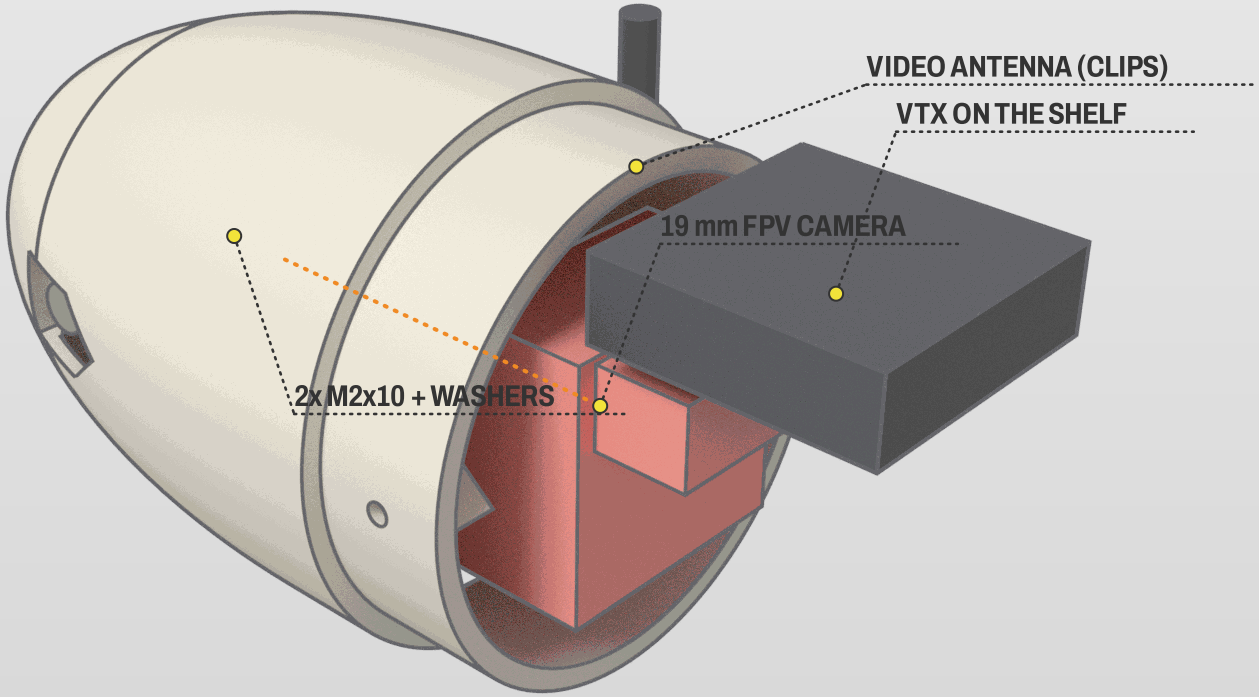
FC on the bay floor on its foam pad, ESC on the FC's top, receiver on the floor ahead of the FC. Battery lead (14 AWG) from the XT30 down the right lane to the FC's power pads; 14 AWG ESC feed from the same pads up the FC's front face onto the ESC; ESC signal (signal, GND, telemetry) from the ESC's front onto the FC's S1 and RX4; receiver lead to the FC's UART6 plug; motor phases low on the left of the wing bay to the MR30, then through the top of the joint. Cut lengths and lanes: Electrical Integration manual. Unplug the MR30 and the servo plugs before pulling the tail module.



HEAT SHRINK

CAMERA NOSE

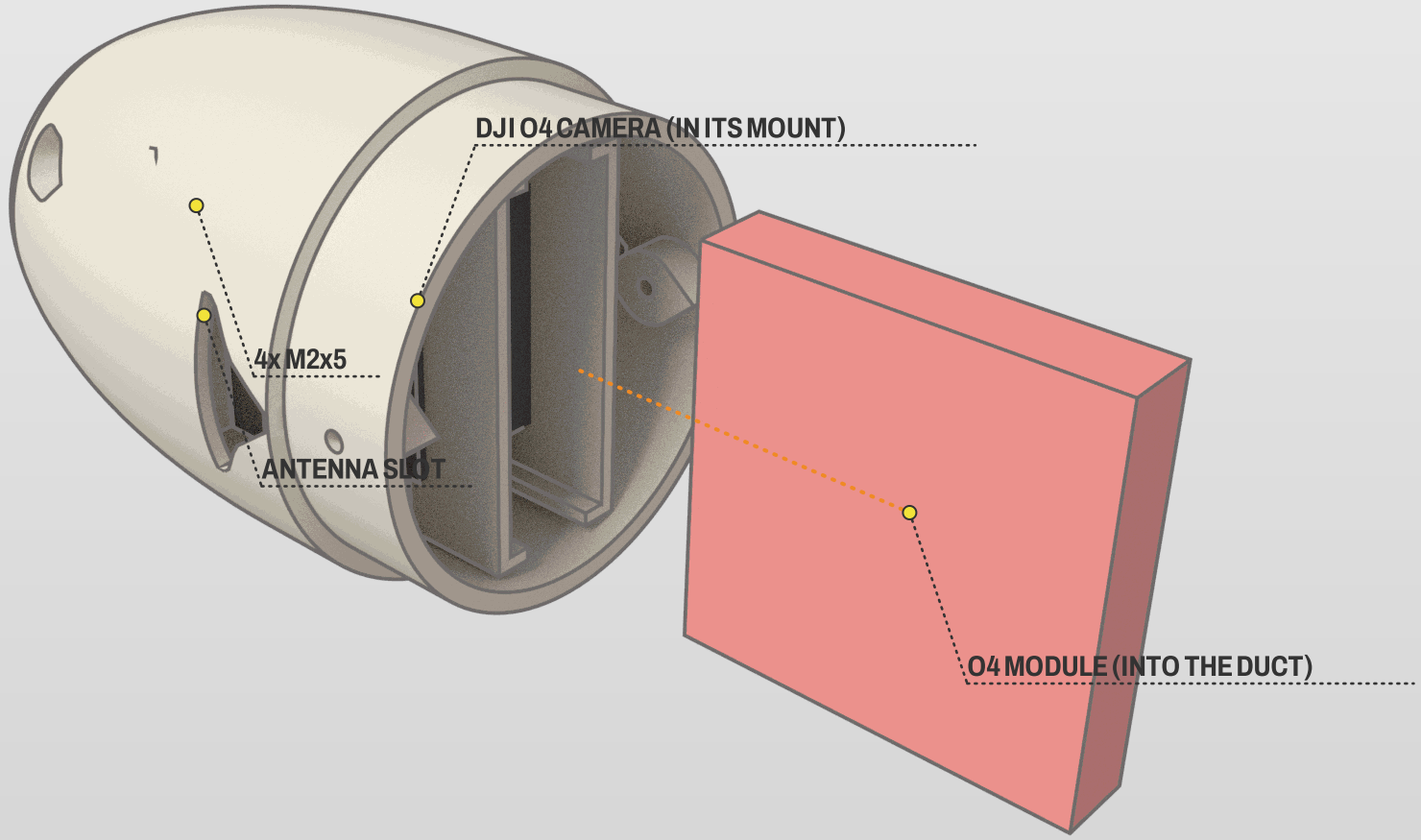
Slide the 19 mm camera in from the open rear until the lens reaches the face; 2 × M2 × 8 pan head + 2 × M2 washer (5 mm OD) through the side seats, lens flush, tilt 0–10° up. Fix the VTX to the printed VTX shelf (foam tape + zip tie through its slots), slide the shelf in from the rear onto its two rails and glue it; video antenna in the two clips on the right wall. Plug the one lead at the nose joint. Dome noses: glue the printed dome ring 3 mm into the front of the tube, then fit the clear dome over its lip with canopy glue (no CA).



FOAM TAPE

CAMERA NOSE

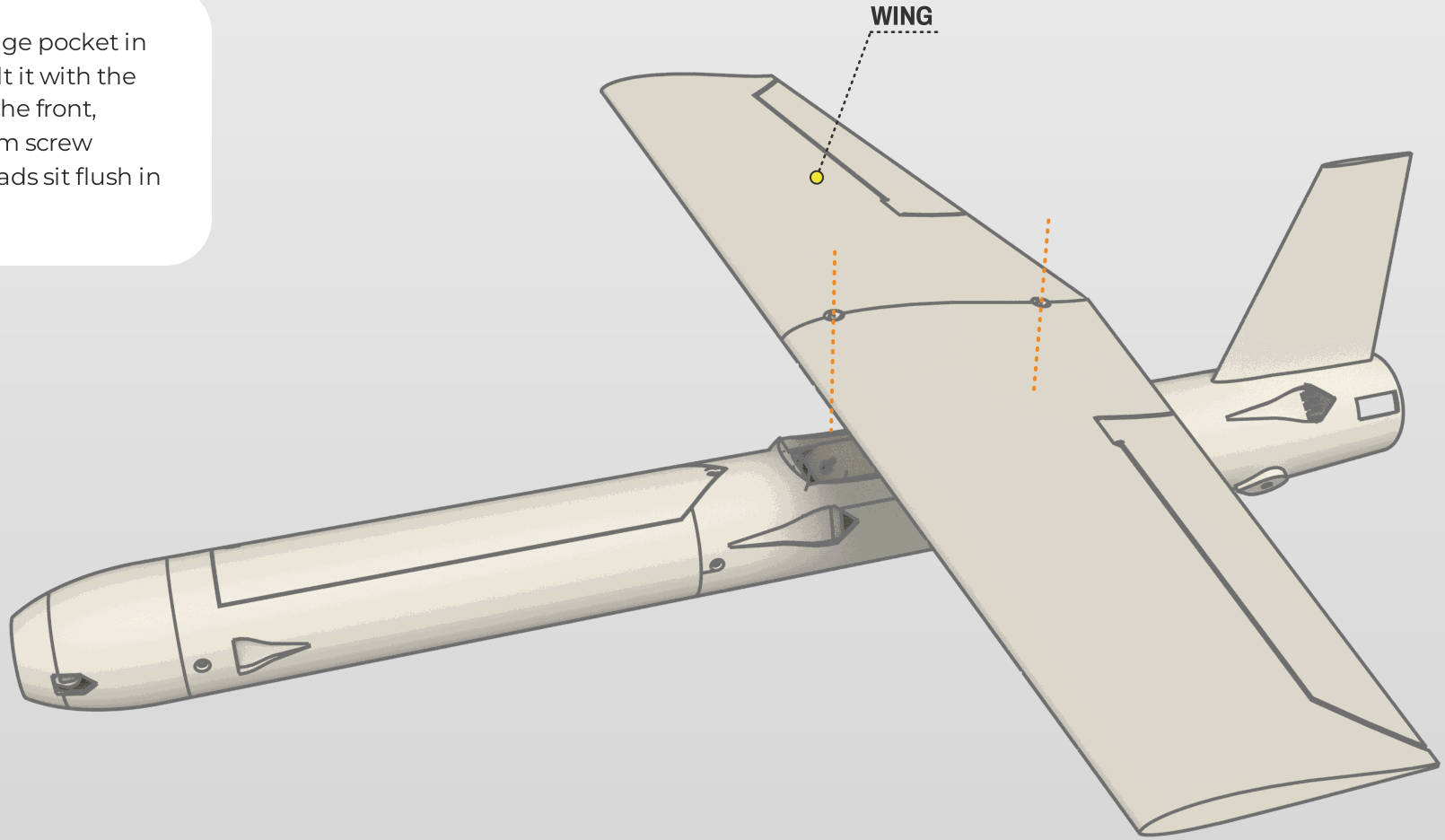
DJI O4 option: clamp the camera in its mount with 4 × M2 × 5 through the side slots (at most 3.1 mm into the mount). Plug the coax and 3-in-1 lead into the module and slide it edge-first into the duct until it stops; foam strip in the top groove, tape over the rear edge, never over its faces.



M2 × 5 SCREWS

FINAL ASSEMBLY

Push an M3 nut up into each bolt-bridge pocket in F2. Set the wing on the saddle and bolt it with the nylon screws: Countersunk M3 × 16 at the front, countersunk M3 × 8 at the rear (a 12 mm screw reaches the aileron servo arm); the heads sit flush in the countersinks.



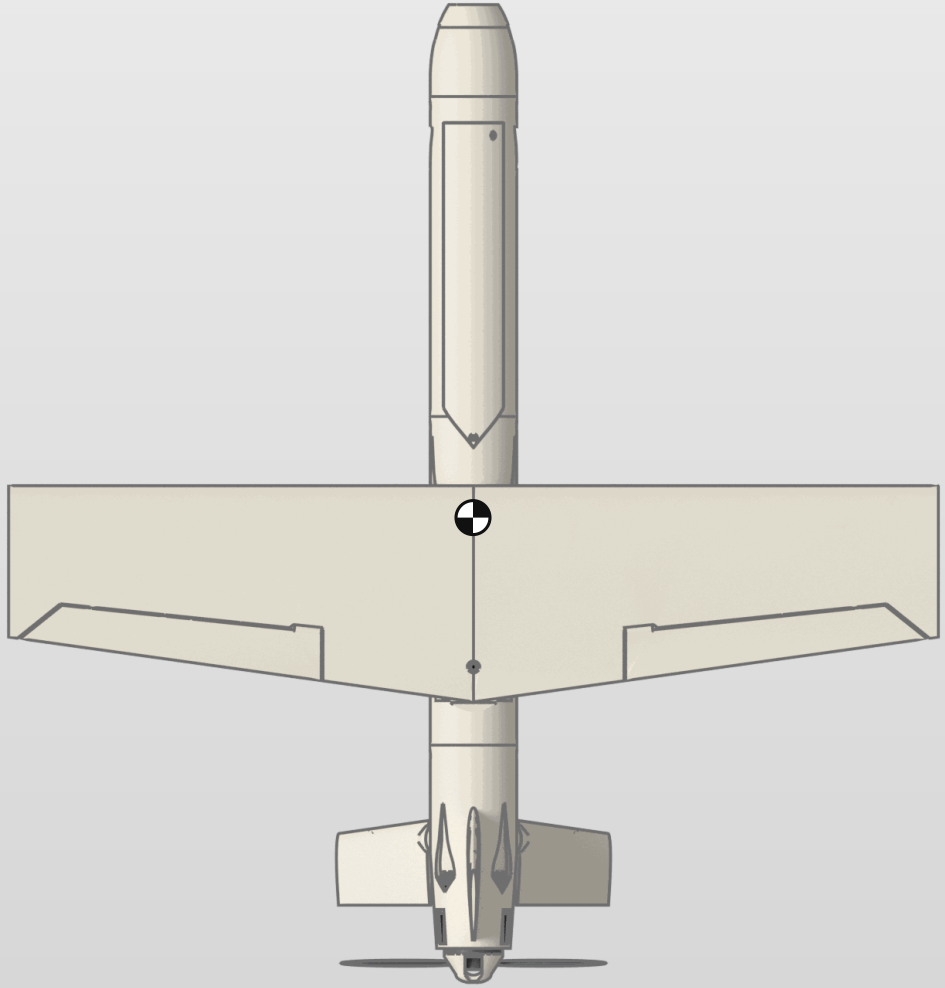
NYLON SCREWS

FINISHING BUILD

Before the first flight: every control surface moves the right way (pitch up: both panel trailing edges up; right roll: right aileron up), throws ±10° on the tail panels, about ±10° (±3.5 mm at the trailing edge) on the ailerons, 30 % expo; the prop turns the right way and pushes aft; failsafe set to motor off.

Balance: CG 15 mm behind the wing leading edge at the root (mark it on the wing underside). Slide the battery to trim; for the first flight set the CG 2 mm forward. If a heavy camera needs it, use the nose or tail ballast cup.

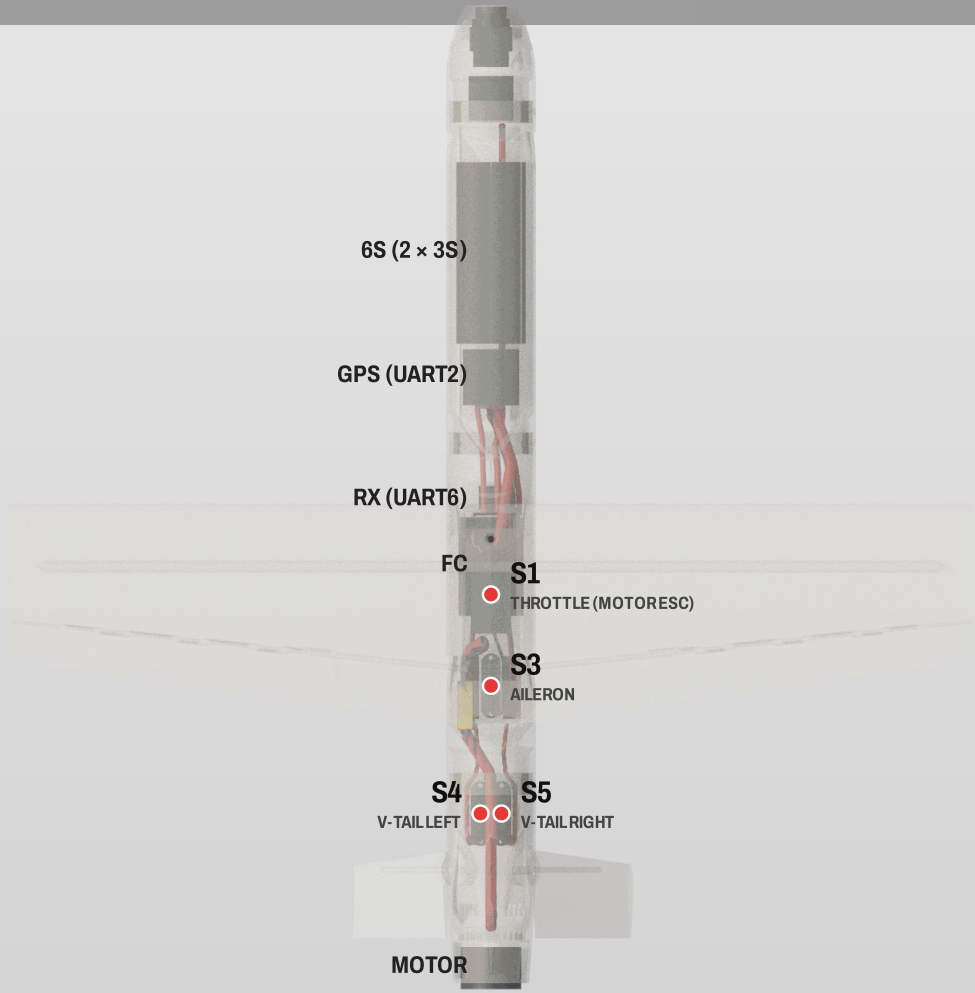
Glass skin: two ±45° layers of 49 g/m² glass on the wing halves and tail panels after printing, then a thin paint finish.



WIRING DIAGRAM

The diagram shows the flight-controller outputs and where each device sits. Signal, power and video leads follow the routes on the wiring page of the assembly guide; the package has this diagram as a separate large page.

PAD	FUNCTION	DEVICE
S1	Throttle (motor ESC)	Sequre SQESC 2670 ESC (bidirectional DShot300)
S2	Unused	Leave empty (DShot group with S1)
S3	Aileron	Aileron servo (ES9052MD, wing bay)
S4	V-tail left	Left tail-panel servo (ES9052MD, tail module)
S5	V-tail right	Right tail-panel servo (ES9052MD, tail module)



ELECTRICAL INTEGRATION

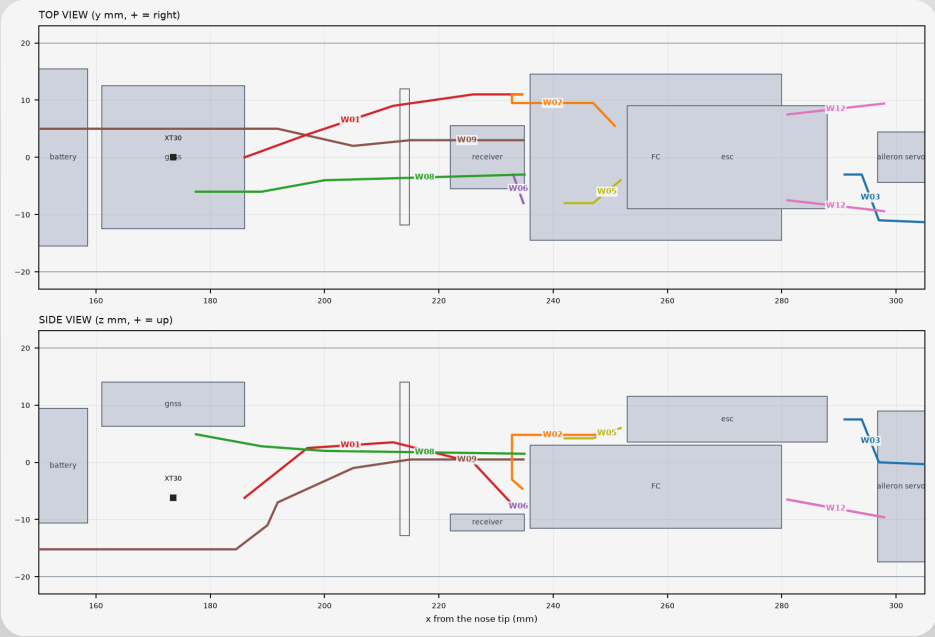
Battery → FC power board (80 A current sensor) → ESC (motor on M1) → MR30 at the tail joint → motor. Main DC path 14 AWG (51 A on a pass, 60 A static), motor phases 16 AWG (42 A RMS). Cable lengths are measured on the 3D harness model; the cut list adds the termination allowances and slack. Full detail, pinouts, settings and tests: the **Electrical Integration manual** (PDF in the package and on the web page).

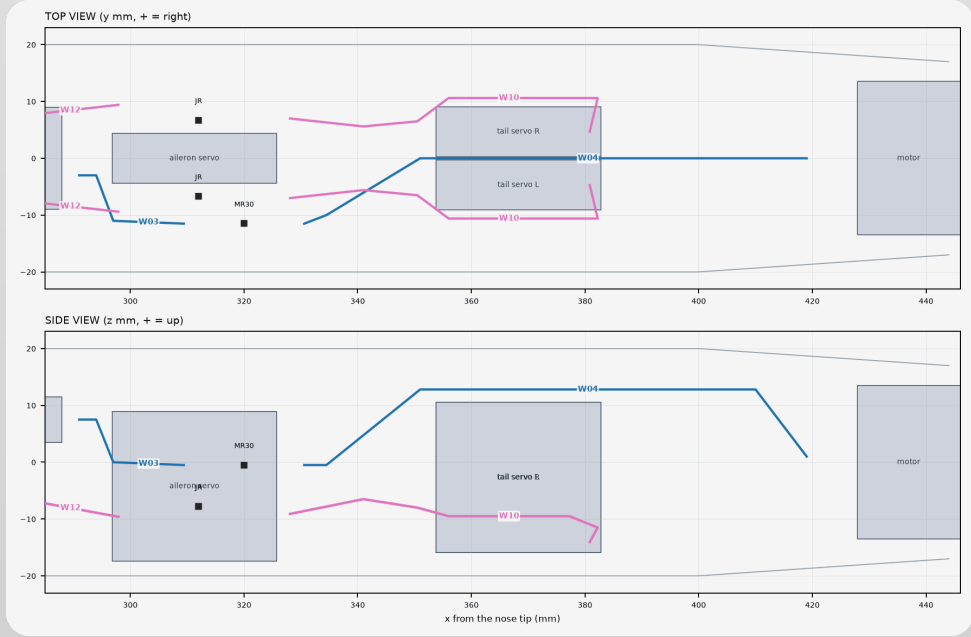
ID	CABLE	FROM	TO	AWG	CUT mm
W01	Battery lead	XT30 female (the pack plugs in), behind the pack under...	Matek H743-WLITE power pads + / - (shared with W02)	14	85
W02	ESC feed	Matek H743-WLITE power pads (after its 90 A sensor)	ESC BAT+ / BAT- (ESC on top of the FC)	14	50
W03	Motor phases, ESC side	Sequre SQESC 2670 motor pads (A, B, C)	MR30 female (ESC side), low on the left ahead of the...	16	50
W04	Motor phases, motor side	MR30 male	Motor phase wires (cut to 20 mm, butt-soldered + heat...	16	145
W05	ESC signal	ESC signal wires (signal, GND, telemetry)	FC S1 (signal, GND); telemetry to RX4	28	40
W06	Receiver lead	RadioMaster RP1 V2 pads (5V, GND, TX, RX), on the bay...	FC UART6 JST-GH (5V, GND, RX6, TX6)	28	35
W08	GPS lead	TBS M10Q (VCC, GND, TX, RX, SDA, SCL)	FC UART2 + I2C1 JST-GH (5V, GND, RX2, TX2, DA1, CL1)	28	95
W09	Video trunk (N0MD nose)	Nose joint plug (camera on the nose; VTX on the nose or,...	FC video pads (power, GND, camera in, VTX out)	28	265
W10	Tail servo leads (L, R)	Emax ES9052MD in F3 (its stock lead)	JR plug pair lying flat beside the aileron servo	26	stock
W12	Tail servo extensions (L, R)	JR plug pair (beside the aileron servo)	FC S4 (left) / S5 (right)	26	stock
W11	Aileron servo lead	Emax ES9052MD lead (stock), aileron servo in the wing bay	FC S3 header	26	stock

ELECTRICAL: POWER AND ROUTING

CABLE	CIRCUIT	CURRENT A (PASS / STATIC)	WIRE	DROP V	dT K / PASS	CONNECTOR RATING
W01+W02	Battery to ESC (DC)	51 / 60	14 AWG	0.07 (0.3 %)	24	XT30: 15 A cont / 30 A peak (see open item)
W03+W04	Motor phases (per phase, RMS)	42 / 49	16 AWG	0.14 (0.6 %)	40	MR30: 30 A cont / 40 A for 1 min per contact

Open item: the XT30 is rated 15 A continuous / 30 A peak, below this aircraft's pass current: an XT60 lead with a ready-made pack that has XT60, or XT30 with full power kept to the short passes. Never connect the ESC lead's VBAT pin to the FC's servo header or a 5 V pad. Power-board jumpers: servo BEC 6 V; Vv battery voltage only if the camera and VTX accept 4S.





ARDUPILOT CONFIGURATION

Starting ArduPlane configuration for SkyG Speed 120 on the Matek H743-WLITE (firmware MatekH743-bdshot, ArduPlane 4.5 or later), with the rpm-governor script skyg_governor.lua. Load the .param file in Mission Planner (Config > Full Parameter List > Load from file), copy the script to the microSD card, reboot, load the .param file again (for the GOV_ parameters), then work through the bench checks.

Flight controller: on the wing-bay floor on a 0.5 mm foam pad, arrow forward; servo rail Vx set to 6 V for the ES9052MD servos.

Firmware: ArduPlane 4.5 or later · **Board:** Matek H743-WLITE (ArduPilot firmware: MatekH743-bdshot)

The setup document and the starting .param file are in the package (ArduPilot folder). Verify every value on the bench before flight.



FLIGHT TEST AND SPEED STAGES

The speed envelope opens in stages. Fly each stage only after the ground checks and the previous stage pass: thrust-stand run, wing load and twist test, control freeplay ≤ 0.2 mm (≤ 0.05 mm above 45 m/s), aileron hinge stiffness, and the tap test. At each run give short aileron and pitch doublets; any buzz or vibration: throttle to zero, gentle pull-up, land and inspect.

Full procedure, limits and log sheet: the test plan in the Engineering folder.

Stage Vne	GOV_RPM (full stick)	Expected level speed
30 m/s	10,400 rpm	30 m/s
35 m/s	12,000 rpm	35 m/s
40 m/s	13,500 rpm	40 m/s
45 m/s	15,000 rpm	44 m/s
50 m/s	16,600 rpm	50 m/s
55 m/s	18,100 rpm	54 m/s
60 m/s	19,700 rpm	60 m/s
65 m/s	21,200 rpm	64 m/s
70 m/s	22,700 rpm	69 m/s
75 m/s	24,300 rpm	74 m/s
80 m/s	25,800 rpm	79 m/s
85 m/s	27,300 rpm	84 m/s
90 m/s	28,800 rpm	89 m/s
95 m/s	30,300 rpm	94 m/s
100 m/s	31,900 rpm	99 m/s
105 m/s	33,400 rpm	104 m/s
110 m/s	34,900 rpm	109 m/s
115 m/s	36,400 rpm	114 m/s
120 m/s	37,900 rpm	118 m/s

SKYG

SKYG SPEED 120

